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Conformal Prediction Under
Distribution Shift: Linking
DRIG and Robust Validation

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Code and data availability

The numerical results in Chapter 4 are based on deterministic population calculations rather than on empirical data. No external or real-world datasets are used.

The Python code used to generate the numerical results and figures implements the population formulas derived in Chapter 3. It computes the fixed-reference coefficient path, the induced residual and score distributions, the total-variation robust-validation radii, the population coverages, and the corresponding interval widths. The code also contains basic numerical consistency checks for the implemented quantities.

The code used to generate the numerical results and figures is available at <https://github.com/cfroidevaux/drig-robust-validation-code>. The repository contains the scripts needed to reproduce the figures and summary values reported in Chapter 4.

Abstract

Conformal prediction provides prediction sets with finite-sample marginal coverage under exchangeability, but this guarantee can fail under distribution shift. Robust validation addresses this problem at the calibration level by enlarging the conformal threshold according to a divergence radius between calibration and test score distributions. Distributional Robustness via Invariant Gradients (DRIG), in contrast, acts at the prediction level by using heterogeneous environments to construct more robust predictors. This thesis studies how these two forms of robustness interact.

The main contribution is a population-level analysis linking a fixed-reference DRIG path to robust validation in a Gaussian structural equation model with a fixed structural matrix. For each value of the DRIG parameter, the analysis derives the induced residual distributions, absolute-residual score laws, total-variation robust-validation radii, coverages, and prediction interval widths. The thesis also distinguishes between risk-level and score-level robustness. The full second-moment uncertainty set associated with the DRIG robustness interpretation gives residual moment control, but it does not by itself control the total-variation radius needed for robust validation.

A numerical population study evaluates the derived quantities in structured Gaussian subclasses and in a setting based on the full DRIG uncertainty set. In the mean-only subclass, increasing the DRIG parameter can align calibration and admissible test score distributions, turning robust-validation intervals from vacuous to finite and informative. In the covariance-only subclass, the same qualitative effect occurs through alignment of residual variances rather than residual means. For the full uncertainty set, robust validation remains vacuous, but Gaussian moment intervals can still become substantially shorter along the fixed-reference path. Overall, the results show that predictor-side robustness and calibration-side robustness are connected through the induced score distributions, but this connection requires score-level structure and cannot be concluded from risk-level robustness alone.

Contents

Notation	ix
1 Introduction	1
2 Theoretical Background	3
2.1 Conformal prediction	3
2.1.1 Prediction sets and marginal coverage	3
2.1.2 Split conformal prediction	4
2.1.3 Exchangeability and score-level calibration	5
2.2 Distribution shift	6
2.2.1 Distribution shift and environment-based modeling	6
2.2.2 Why standard conformal prediction can fail under distribution shift	7
2.3 Robust validation	7
2.4 Distributional Robustness via Invariant Gradients	10
2.4.1 Robust prediction across environments	10
2.4.2 The DRIG objective and a fixed-reference specialization	11
2.4.3 Robustness interpretation and risk-level uncertainty sets	12
2.5 Combining robust prediction and robust calibration	14
3 A Worked Population Analysis	15
3.1 A fixed- B Gaussian SCM	15
3.2 The fixed-reference DRIG path	17
3.2.1 Residual and risk representation	17
3.2.2 Reference and pooled OLS	17
3.2.3 Fixed-reference objective and coefficient path	18
3.3 Residual and score distributions along the DRIG path	19
3.4 Population conformal prediction and robust validation	20
3.4.1 Standard conformal intervals	21
3.4.2 Robust-validation intervals	21
3.5 Risk-level robustness and score-level limitations	23
3.6 Structured Gaussian subclasses	25
3.6.1 Mean-only Gaussian shifts	25
3.6.2 Covariance-only Gaussian shifts	26
4 Numerical Population Study	29
4.1 Setup and evaluation criteria	29
4.2 Mean-only Gaussian subclass	31
4.3 Covariance-only Gaussian subclass	34
4.4 Full second-moment uncertainty set	36
4.5 Summary	38
5 Discussion	41
Bibliography	45
A Proofs	47
A.1 Proof of Proposition 3.1	47

A.2	Proof of Proposition 3.2	48
A.3	Proof of Proposition 3.6	49
A.4	Proof of Proposition 3.7	50
A.5	Proof of Lemma 3.8	50
A.6	Proof of Proposition 3.9	51
A.7	Proof of Proposition 3.10	54

Notation

This section summarizes notation used repeatedly throughout the thesis. Further notation is introduced locally when needed.

General notation.

$\mathcal{L}(X)$	Distribution of a random variable X .
$\ P - Q\ _{\text{TV}}$	Total variation distance between probability distributions P and Q .
$A \preceq B$	Loewner order for symmetric matrices, meaning that $B - A$ is positive semidefinite.
Φ, ϕ, z_τ	Standard normal distribution function, density, and τ -quantile $z_\tau := \Phi^{-1}(\tau)$.

Conformal prediction and robust validation.

$\mathcal{X}, \mathcal{Y}$	Feature and response spaces.
$s(x, y), S$	Conformal score and corresponding random score $S = s(X, Y)$.
Q, P	Joint distributions on $\mathcal{X} \times \mathcal{Y}$ are denoted by Q . Score distributions are typically denoted by P . Future noise laws are denoted by P_v .
$P_{\text{cal}}, P_{\text{test}}$	Calibration and test score distributions.
α	Target miscoverage level. The nominal coverage level is $1 - \alpha$.
ρ	Total-variation robust-validation radius. For a single test score law, $\rho = 2\ P_{\text{test}} - P_{\text{cal}}\ _{\text{TV}}$.

Environments and structural model.

$\mathcal{E}_{\text{obs}}, e, \omega^e$	Set of observed environments, environment index, and environment weights with $\sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e = 1$.
$t, \mathcal{A}_\gamma$	Structured test shifts may be indexed by t . Admissible classes of future noise laws are denoted by $\mathcal{A}_\gamma$.
Z^e	Covariate-response vector $Z^e = ((X^e)^\top, Y^e)^\top$ in environment e .
B, K	Fixed structural matrix and inverse structural matrix $K = (I - B)^{-1}$.
$\xi^e, \delta^e, \Omega_e$	SCM noise vector in environment e , with $\xi^e \sim N(\delta^e, \Omega_e)$.
Λ_e, S_e	Second-moment matrices

$$\Lambda_e := \mathbb{E}[Z^e(Z^e)^\top], \quad S_e := \mathbb{E}[\xi^e(\xi^e)^\top] = \Omega_e + \delta^e(\delta^e)^\top.$$

$\Lambda_{XX}^e, \Lambda_{XY}^e, \Lambda_{YY}^e$	Blocks of Λ_e corresponding to the covariate and response coordinates.
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DRIG path, residuals, and scores.

b, b_γ	Linear predictor coefficient and fixed-reference DRIG coefficient at parameter γ . The predictor is $x \mapsto b^\top x$.
Γ	Admissible parameter set for the coefficient path, $\Gamma := \{\gamma \geq 0 : A_\gamma \succ 0\}$.
ℓ_b, w_b, w_γ	Residual directions

$$\ell_b := \begin{pmatrix} -b \\ 1 \end{pmatrix}, \quad w_b := K^\top \ell_b, \quad w_\gamma := w_{b_\gamma}.$$

$R_e(b)$	Squared prediction risk $R_e(b) := \mathbb{E}[(Y^e - b^\top X^e)^2]$.
U_γ^e, T_γ^e	Residual and absolute residual score under b_γ , $U_\gamma^e = Y^e - b_\gamma^\top X^e$ and $T_\gamma^e = U_\gamma^e $.
$F_{e,\gamma}, f_{e,\gamma}$	Distribution function and density of T_γ^e .
$F_{\text{cal},\gamma}, T_{\text{cal},\gamma}, P_{\text{cal},\gamma}^T$	Weighted calibration score law, a random variable with this law, and distribution notation $P_{\text{cal},\gamma}^T = \mathcal{L}(T_{\text{cal},\gamma})$.
$F_{t,\gamma}$	Score law under a structured test shift t .
$P_{v,\gamma}^T$	Score law induced by a future noise law P_v , $P_{v,\gamma}^T := \mathcal{L}(w_\gamma^\top v)$, where $v \sim P_v$.

Uncertainty sets and population quantities.

$S_\gamma, \mathcal{C}_\gamma$	Risk-level second-moment matrix and uncertainty set
$S_\gamma := S_0 + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (S_e - S_0), \quad \mathcal{C}_\gamma := \{P_v : \mathbb{E}[vv^\top] \preceq S_\gamma\}.$	
$\mathcal{C}_\gamma^{\text{mean},G}, \mathcal{C}_\gamma^{\text{cov},G}$	Structured Gaussian mean-only and covariance-only subclasses.
$M_{\text{obs}}, \Delta_{\text{obs}}$	Weighted observed mean-shift second moment and weighted observed covariance perturbation.
M_γ	Residual second-moment bound $M_\gamma := w_\gamma^\top S_\gamma w_\gamma$.
$q_\gamma^{\text{std}}, q_\gamma^{\text{RV}}$	Population thresholds for standard conformal prediction and robust validation, evaluated at miscoverage level α . The admissible class is suppressed when clear from context.
$\text{Width}_\gamma^{\text{std}}, \text{Width}_\gamma^{\text{RV}}$	Standard conformal and robust-validation interval widths.
$\text{Cov}_{t,\gamma}^{\text{std}}, \text{Cov}_{t,\gamma}^{\text{RV}}$	Standard conformal and robust-validation population coverages under test shift t .
$\rho_t(\gamma), \rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma)$	Radius for a single test shift and worst-case radius over an admissible class, which is abbreviated as $\rho_{\text{wc}}(\gamma)$ when the class is clear.
η, S_η	Fixed-target severity and, in the full second-moment setting, the corresponding fixed target matrix.
$\gamma_{\text{fin}}, \gamma_{\text{width}}^*$	First grid value with finite matched robust-validation width and grid-level minimizer of finite matched robust-validation width.
$\gamma_{\text{risk},\eta}^*, \gamma_{\text{width},\eta}^*$	Grid-level minimizers of fixed-target residual second moment and fixed-target robust-validation width.

Chapter 1

Introduction

Modern prediction methods are often evaluated by their point prediction accuracy. In many applications, however, a point prediction alone is not enough. A prediction should also quantify its uncertainty, especially when future data may differ from the data used for model development. Conformal prediction provides a general way to construct prediction sets with finite-sample marginal coverage under exchangeability ([Angelopoulos et al., 2026](#)). Its split version is particularly attractive because it can be combined with almost any fitted prediction model: a calibration sample is used to compute an empirical quantile of nonconformity scores, and this quantile determines the prediction set. The resulting coverage guarantee does not require the fitted predictor to be correctly specified or highly accurate. It relies on exchangeability of the calibration and test scores.

The coverage guarantee of standard split conformal prediction therefore rests on a symmetry condition. Conditional on the fitted prediction rule, the calibration scores and the future test score must be exchangeable. This condition is satisfied in the usual setting where the calibration and test data are drawn independently from the same distribution, but it is no longer justified under distribution shift. If the test score distribution differs from the calibration score distribution, then a calibration quantile computed from the observed scores can be too small for future test points. In regression with absolute residual scores, this means that prediction intervals calibrated on one environment may undercover in another environment.

One response to this problem is robust validation. Robust validation modifies the calibration step rather than the fitted prediction rule. It assumes that the future test score distribution lies in an uncertainty set around the calibration score distribution and chooses a more conservative score threshold to maintain coverage ([Cauchois et al., 2024](#)). In the total-variation case, the adjustment has a particularly simple form: the quantile level is increased by one half of the radius of the score-distribution shift, with clipping at one. A larger radius protects against larger shifts, but it also leads to wider prediction sets and can make the resulting intervals uninformative.

A complementary response is to change the predictor itself. Distributional Robustness via Invariant Gradients (DRIG) uses heterogeneous environments to construct predictors that are robust against structured distribution shifts ([Shen et al., 2026](#)). In contrast to robust validation, DRIG acts before calibration: it changes the regression coefficient and hence changes the residuals. Since conformal prediction with absolute residual scores calibrates these residuals, changing the predictor can also change the calibration and test

score distributions.

This thesis studies this connection at the population level. The guiding question is whether changing the predictor through a DRIG-type path can reduce the score-distribution shift that robust validation must guard against. Equivalently, the thesis asks how the DRIG tuning parameter affects the robust-validation radius and the resulting prediction interval width. This question is not answered by prediction risk alone. A predictor with favourable squared-error risk need not minimize the width of robust conformal prediction intervals, because the interval width depends on the calibration score distribution and on its discrepancy from the test score distribution.

The main contribution is a worked population analysis in a Gaussian structural equation model with a fixed structural matrix. The analysis uses a fixed-reference specialization of the DRIG objective. This specialization is not the full minimum-risk DRIG objective, but it yields an explicit quadratic path of population predictors and makes the interaction with robust validation analytically tractable. For each value of the tuning parameter, the thesis derives the corresponding residual distributions, absolute-residual score laws, total-variation robust-validation radii, coverages, and interval widths.

The analysis also clarifies a limitation of using the full DRIG uncertainty set directly for robust validation. This uncertainty set gives residual moment control and is therefore natural for squared prediction risk. Robust validation, however, requires control of the discrepancy between score distributions. The thesis shows that the full uncertainty set alone does not control the total-variation radius of the induced score laws. It then identifies structured Gaussian subclasses in which the DRIG penalty controls a one-dimensional feature of the residual law, such as its mean or variance, that is relevant for the conformal score distribution.

The numerical population study illustrates both sides of this distinction. In a mean-only Gaussian subclass, increasing the DRIG parameter can align calibration and admissible test score distributions so that robust-validation intervals become finite and informative. In a covariance-only Gaussian subclass, the same qualitative effect occurs through alignment of residual variances rather than residual means. In a setting based on the full DRIG uncertainty set, robust validation remains vacuous, but Gaussian moment intervals can still become substantially shorter along the fixed-reference path.

The remainder of the thesis is organized as follows. Chapter 2 introduces the theoretical background on conformal prediction, distribution shift, robust validation, and DRIG. Chapter 3 develops the worked population analysis and derives the score laws and robust-validation quantities along the fixed-reference DRIG path. Chapter 4 evaluates these quantities numerically in structured Gaussian subclasses and in a setting based on the full DRIG uncertainty set. Chapter 5 discusses the main findings, limitations, and possible extensions.

Chapter 2

Theoretical Background

This chapter introduces the theoretical concepts used in the worked population analysis in Chapter 3 and in the numerical population study in Chapter 4. The presentation focuses on conformal score distributions, robust validation, and DRIG, because the thesis studies how predictor-side robustness affects calibration-side robustness under distribution shift.

2.1 Conformal prediction

2.1.1 Prediction sets and marginal coverage

In many prediction problems, a point prediction alone is not sufficient. A predictor that returns only a single value does not quantify how uncertain the prediction is and does not distinguish between easy and difficult test points. In regression, uncertainty is often represented by a prediction interval. More generally, for a feature value $x \in \mathcal{X}$, one seeks a prediction set $C(x) \subseteq \mathcal{Y}$ that contains plausible values of the response.

Conformal prediction provides a general framework for constructing such prediction sets from data and a conformal score function. Its main appeal is that it can be combined with arbitrary prediction methods and yields finite-sample coverage guarantees under exchangeability, without requiring a correctly specified parametric model for the data-generating distribution (Angelopoulos and Bates, 2022).

For a prescribed miscoverage level $\alpha \in (0, 1)$, the target guarantee is

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha. \quad (2.1)$$

This property is called *marginal coverage*, because the guarantee is marginal over the feature value X_{n+1} . That is, it averages over the distribution of X_{n+1} rather than requiring the same coverage level at every fixed $x \in \mathcal{X}$. The probability in (2.1) is also taken with respect to the random data used to construct the random set C . It is not a statement conditional on a fixed feature value $X_{n+1} = x$. In particular, marginal coverage does not imply that

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha$$

holds for every $x \in \mathcal{X}$ (Angelopoulos et al., 2026, Section 4.3).

This thesis focuses on split conformal prediction for regression, introduced next, because it separates model fitting from calibration.

2.1.2 Split conformal prediction

Split conformal prediction uses two separate sources of data. The formulation below follows [Angelopoulos et al. \(2026, Definition 3.5 and Algorithm 3.6\)](#). First, let $\mathcal{D}_{\text{train}}$ denote a training dataset used to fit a prediction rule or, more generally, to construct a score function. In regression, a fitted prediction rule $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$ can for instance be turned into the absolute residual score $s(x, y) = |y - \hat{f}(x)|$. Second, let $\mathcal{D}_n := ((X_1, Y_1), \dots, (X_n, Y_n))$ denote the calibration sample, and let (X_{n+1}, Y_{n+1}) denote a future test point. The calibration sample is not used to fit the prediction rule, it is used only to calibrate a score threshold.

A *conformal score* is a measurable function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, where larger values indicate worse agreement between a candidate response y and the fitted prediction rule at feature value x . The score function may depend on $\mathcal{D}_{\text{train}}$, but after conditioning on $\mathcal{D}_{\text{train}}$ it is treated as fixed. The calibration scores are $S_i := s(X_i, Y_i)$, $i = 1, \dots, n$. Let

$$\hat{F}_n(t) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{S_i \leq t\}$$

be the empirical distribution function of the calibration scores. For all $\beta > 0$, define the empirical upper quantile by

$$\hat{Q}_n(\beta) := \inf\{t \in \mathbb{R} : \hat{F}_n(t) \geq \beta\},$$

with the convention that $\inf \emptyset = +\infty$. This convention matters when the requested quantile level is larger than one.

For a miscoverage level $\alpha \in (0, 1)$, split conformal prediction uses the finite-sample corrected level $\beta_n(\alpha) := \lceil (n+1)(1-\alpha) \rceil / n$ and defines

$$\hat{q}_{1-\alpha} := \hat{Q}_n(\beta_n(\alpha)). \quad (2.2)$$

The split conformal prediction set at a new feature value x is

$$C(x) := \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}_{1-\alpha}\}. \quad (2.3)$$

If $\beta_n(\alpha) > 1$, then $\hat{q}_{1-\alpha} = +\infty$, and the resulting prediction set is trivially valid but uninformative. Thus, split conformal prediction includes exactly those response values whose score does not exceed the calibrated threshold ([Angelopoulos et al., 2026, Algorithm 3.6](#)).

The following result states the finite-sample coverage guarantee of split conformal prediction under exchangeability.

Theorem 2.1 (Split conformal marginal coverage). *Assume that, conditional on $\mathcal{D}_{\text{train}}$, the calibration sample $(X_1, Y_1), \dots, (X_n, Y_n)$ and the test point (X_{n+1}, Y_{n+1}) are exchangeable. Then the split conformal prediction set in (2.3) satisfies*

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

A formal proof can be found in [Angelopoulos et al. \(2026, Theorem 3.2\)](#). The proof relies on exchangeability of the calibration scores and the future test score: conditional on the fitted score function, the rank of $s(X_{n+1}, Y_{n+1})$ among the calibration scores and the future test score is uniformly distributed up to ties, as discussed in Section 2.1.3.

In the regression problems considered in this thesis, the score is the absolute residual score $s(x, y) = |y - \hat{f}(x)|$. With this score, (2.3) becomes the interval

$$C(x) = [\hat{f}(x) - \hat{q}_{1-\alpha}, \hat{f}(x) + \hat{q}_{1-\alpha}].$$

Hence, the width of the prediction interval is determined by an empirical upper quantile of the calibration residuals.

2.1.3 Exchangeability and score-level calibration

The finite-sample validity of split conformal prediction is based on exchangeability.

Definition 2.2 (Exchangeability). *A random vector $(Z_1, \dots, Z_m)$ with values in a measurable space $\mathcal{Z}$ is called exchangeable if, for every permutation σ of $\{1, \dots, m\}$,*

$$(Z_1, \dots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, \dots, Z_{\sigma(m)}).$$

Here, $\stackrel{d}{=}$ denotes equality in distribution. Independent and identically distributed (i.i.d.) random variables are exchangeable, but exchangeability is broader and also allows certain forms of dependence (Angelopoulos et al., 2026, Definition 2.1).

In split conformal prediction, the relevant exchangeability is exchangeability of the calibration and test scores. To see why, condition on the training data $\mathcal{D}_{\text{train}}$. Then the score function s is fixed. If $(X_1, Y_1), \dots, (X_n, Y_n), (X_{n+1}, Y_{n+1})$ are exchangeable and are independent of $\mathcal{D}_{\text{train}}$, then the score vector

$$(S_1, \dots, S_n, S_{n+1}), \quad S_i := s(X_i, Y_i), \quad i = 1, \dots, n+1,$$

is exchangeable conditional on $\mathcal{D}_{\text{train}}$. This follows because applying the same fixed measurable function to each coordinate preserves exchangeability (Angelopoulos et al., 2026, Section 3.4.1).

This score-level exchangeability is the mechanism behind Theorem 2.1. Once the score function is fixed, the calibration scores and the future test score are treated symmetrically. The future score is therefore calibrated against scores with the same exchangeable structure. The finite-sample correction in (2.2) accounts for the fact that the threshold is computed from n calibration scores but must cover one additional test score. It ensures that

$$\mathbb{P}(S_{n+1} \leq \hat{q}_{1-\alpha}) \geq 1 - \alpha,$$

which is equivalent to the marginal coverage statement because $S_{n+1} \leq \hat{q}_{1-\alpha}$ holds exactly when $Y_{n+1} \in C(X_{n+1})$. The conditioning argument and the reduction from exchangeability of data points to exchangeability of scores are discussed in Angelopoulos et al. (2026, Section 3.4.1).

The important conclusion for the rest of the thesis is that split conformal prediction calibrates a threshold on the score distribution. Under exchangeability, the calibration scores and the test score have the symmetry needed for this threshold to be valid. Under distribution shift, this symmetry may fail. The next section therefore studies distribution shift through its effect on calibration and test score distributions.

2.2 Distribution shift

2.2.1 Distribution shift and environment-based modeling

Distribution shift describes situations in which the future test point is generated under a different distribution than the calibration data. Let Q_{cal} denote the joint distribution on $\mathcal{X} \times \mathcal{Y}$ that generates the calibration data, and let Q_{test} denote the joint distribution that generates the future test point. In this thesis, *distribution shift* means that

$$Q_{\text{test}} \neq Q_{\text{cal}}.$$

This definition is deliberately broad. It includes, for example, changes in the marginal distribution of the covariates, changes in the conditional response mechanism, and combinations of both. Conformal methods under such shifts are discussed, for instance, in [Angelopoulos et al. \(2026, Section 7.2\)](#).

For conformal prediction, the relevant distributional object is the distribution of the score. Fix a score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, after conditioning on the training data $\mathcal{D}_{\text{train}}$. The score function induces a distribution on the real line from any joint distribution Q on $\mathcal{X} \times \mathcal{Y}$. We write $P := Q \circ s^{-1}$ for this induced score distribution, that is,

$$P(B) = Q(\{(x, y) \in \mathcal{X} \times \mathcal{Y} : s(x, y) \in B\})$$

for every Borel set $B \subseteq \mathbb{R}$. In particular,

$$P_{\text{cal}} := Q_{\text{cal}} \circ s^{-1}, \quad P_{\text{test}} := Q_{\text{test}} \circ s^{-1}$$

denote the calibration and test score distributions.

The distinction between joint distributions and score distributions is important. A shift in Q matters for split conformal prediction because it may change the induced score distribution P . It is possible that $Q_{\text{cal}} \neq Q_{\text{test}}$ while the induced score distributions are still equal for a particular score function. Conversely, if $P_{\text{cal}} \neq P_{\text{test}}$, then the calibration scores and the test score no longer have the same marginal distribution. Since exchangeability implies equality of marginal distributions, the score-level exchangeability argument from [Section 2.1.3](#) cannot apply.

A convenient way to describe distribution shift is to index different data-generating regimes by environments. Let $\mathcal{E}$ be a finite set of environments. Each environment $e \in \mathcal{E}$ corresponds to a joint distribution Q^e on $\mathcal{X} \times \mathcal{Y}$.

Observed environments represent the regimes available during model development or calibration, whereas test environments represent possible future regimes. This environment-based perspective is natural in robust prediction methods that use heterogeneity across observed data sources to prepare for future distribution shifts, such as anchor regression and DRIG ([Rothenhäusler et al., 2021](#); [Shen et al., 2026](#)).

In later chapters, environments will be specified more concretely through a fixed- B linear structural representation with Gaussian noise laws. At this point, however, only the induced distributions Q^e and the corresponding score distributions are needed. The distinction between joint distributions and score distributions is important because conformal validity depends on calibration at the level of scores, while robust prediction methods often model heterogeneity at the level of environments or joint distributions.

2.2.2 Why standard conformal prediction can fail under distribution shift

Consider again a fixed score function s , conditional on the training data $\mathcal{D}_{\text{train}}$. Let $S_i := s(X_i, Y_i)$, $i = 1, \dots, n$, be the calibration scores, and let $S_{n+1} := s(X_{n+1}, Y_{n+1})$ be the future test score. If $(X_i, Y_i) \sim Q_{\text{cal}}$, $i = 1, \dots, n$, whereas $(X_{n+1}, Y_{n+1}) \sim Q_{\text{test}}$, then $S_i \sim P_{\text{cal}}$, $i = 1, \dots, n$, whereas $S_{n+1} \sim P_{\text{test}}$.

If the score vector $(S_1, \dots, S_n, S_{n+1})$ were exchangeable, then all entries would have the same marginal distribution. Hence exchangeability of the score vector implies $P_{\text{cal}} = P_{\text{test}}$. Therefore, if $P_{\text{cal}} \neq P_{\text{test}}$, the calibration scores and the test score cannot be exchangeable. Equality of the marginal score distributions alone would not imply exchangeability, but its failure is already enough to break the standard conformal argument: the calibration quantile is computed from scores drawn from P_{cal} , while coverage is evaluated under P_{test} .

The population analogue makes the issue transparent. Let

$$q_{\text{cal}}(1 - \alpha) := \inf\{q \in \mathbb{R} : P_{\text{cal}}(S \leq q) \geq 1 - \alpha\}$$

be the $(1 - \alpha)$ -quantile of the calibration score distribution. If this threshold is used for a test point with score distribution P_{test} , the resulting population coverage is

$$P_{\text{test}}(S \leq q_{\text{cal}}(1 - \alpha)),$$

which need not be at least $1 - \alpha$. In particular, if the test scores tend to be larger than the calibration scores, then the calibration quantile can be too small and the resulting prediction sets can undercover.

For absolute residual scores $s(x, y) = |y - \hat{f}(x)|$, this failure has a direct interpretation. If the predictor has larger residuals under the test distribution than under the calibration distribution, then the calibration residual quantile may underestimate the residual size needed for the test environment. Standard split conformal prediction then produces intervals that are too narrow for the future test data.

The rest of the thesis uses this score-level view. Robust validation controls coverage by modifying the calibration threshold, while DRIG changes the predictor and hence the induced residual and score distributions.

2.3 Robust validation

Robust validation addresses score-distribution shift at the calibration stage. It keeps the predictor and the score function fixed, but chooses a more conservative score threshold that remains valid for a specified class of possible test score distributions (Cauchois et al., 2024).

Throughout this section, the score function is fixed. Let P_{cal} denote the calibration score distribution. Robust validation assumes that the future test score distribution belongs to an uncertainty set around P_{cal} . Thus, the uncertainty set is placed directly on the induced score distributions. This is the relevant level for conformal coverage.

Following Cauchois et al. (2024, Section 2.1), robust validation defines the uncertainty set through an f -divergence. To avoid overloading notation with the predictor, we write the divergence generator as φ .

Definition 2.3 (φ -divergence). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R} \cup \{\infty\}$ be a closed convex function satisfying $\varphi(1) = 0$ and $\varphi(t) = \infty$ for all $t < 0$. We denote the corresponding φ -divergence between two score distributions P and P_0 by $D_\varphi(P\|P_0)$. If P is absolutely continuous with respect to P_0 , then*

$$D_\varphi(P\|P_0) := \int \varphi\left(\frac{dP}{dP_0}\right) dP_0.$$

For a radius $\rho \geq 0$, define the score-distribution uncertainty set

$$\mathcal{P}_{\varphi,\rho}(P_{\text{cal}}) := \{P : D_\varphi(P\|P_{\text{cal}}) \leq \rho\}.$$

The radius ρ specifies how much score-distribution shift the method is designed to tolerate.

For all $\beta \in [0, 1]$ and all score distributions P , define the population quantile

$$Q_P(\beta) := \inf\{q \in \mathbb{R} : P((-\infty, q]) \geq \beta\}.$$

Equivalently, if $S \sim P$, then $Q_P(\beta) = \inf\{q \in \mathbb{R} : \mathbb{P}(S \leq q) \geq \beta\}$. We use the convention that this quantile may be $+\infty$, for example when $\beta = 1$ and the score distribution is unbounded above.

The population robust-validation threshold is the worst-case upper quantile over all score distributions in the uncertainty set:

$$Q_{\varphi,\rho}^{\text{WC}}(1 - \alpha; P_{\text{cal}}) := \sup_{P \in \mathcal{P}_{\varphi,\rho}(P_{\text{cal}})} Q_P(1 - \alpha). \quad (2.4)$$

The corresponding population robust-validation prediction set is

$$C_{\varphi,\rho}(x; P_{\text{cal}}) := \left\{y \in \mathcal{Y} : s(x, y) \leq Q_{\varphi,\rho}^{\text{WC}}(1 - \alpha; P_{\text{cal}})\right\}.$$

This construction gives a direct population guarantee. If the test score distribution P_{test} satisfies $D_\varphi(P_{\text{test}}\|P_{\text{cal}}) \leq \rho$, then $P_{\text{test}} \in \mathcal{P}_{\varphi,\rho}(P_{\text{cal}})$. By the definition of the worst-case quantile in (2.4), if $S \sim P_{\text{test}}$, then

$$\mathbb{P}(S \leq Q_{\varphi,\rho}^{\text{WC}}(1 - \alpha; P_{\text{cal}})) \geq 1 - \alpha.$$

Equivalently, for a future test point (X, Y) with induced score distribution P_{test} ,

$$\mathbb{P}(Y \in C_{\varphi,\rho}(X; P_{\text{cal}})) \geq 1 - \alpha.$$

Thus, robust validation protects coverage by replacing the usual calibration quantile with a worst-case quantile over plausible test score distributions.

A useful simplification is that the worst-case quantile can be represented as an ordinary quantile of the calibration score distribution at a more conservative level.

Theorem 2.4 (Robust-validation quantile representation). *For the robust-validation uncertainty set $\mathcal{P}_{\varphi,\rho}(P_{\text{cal}})$, there exists a function $g_{\varphi,\rho} : [0, 1] \rightarrow [0, 1]$, depending only on φ and ρ , whose generalized inverse satisfies*

$$Q_{\varphi,\rho}^{\text{WC}}(1 - \alpha; P_{\text{cal}}) = Q_{P_{\text{cal}}}(g_{\varphi,\rho}^{-1}(1 - \alpha)).$$

This representation is due to [Cauchois et al. \(2024, Proposition 1\)](#). Hence, robust validation can be interpreted as increasing the quantile level from $1 - \alpha$ to the robustness-adjusted level $g_{\varphi, \rho}^{-1}(1 - \alpha)$.

In Chapter 3, this thesis uses the total-variation case, which is one particular f -divergence.

Definition 2.5 (Total variation distance). *For probability distributions P and P_0 on $\mathbb{R}$, the total variation distance is*

$$\|P - P_0\|_{\text{TV}} := \sup_{B \in \mathcal{B}(\mathbb{R})} |P(B) - P_0(B)|,$$

where $\mathcal{B}(\mathbb{R})$ denotes the Borel sets on $\mathbb{R}$.

If P and P_0 have densities p and p_0 with respect to a common dominating measure, then equivalently

$$2\|P - P_0\|_{\text{TV}} = \int |p(z) - p_0(z)| dz.$$

In the rest of the thesis, the total-variation radius is used directly through this metric definition. Unlike the likelihood-ratio expression in Definition 2.3, this definition is well-defined for arbitrary score distributions and does not require absolute continuity.

The total-variation case corresponds to $\varphi(t) = |t - 1|$, for which $D_\varphi(P \| P_0) = 2\|P - P_0\|_{\text{TV}}$. By [Cauchois et al. \(2024, Example 1\)](#), in this case,

$$g_{\varphi, \rho}^{-1}(\tau) = \min \left\{ \tau + \frac{\rho}{2}, 1 \right\},$$

and therefore the robust quantile level is

$$u_\rho^{\text{TV}}(\alpha) := \min \left\{ 1 - \alpha + \frac{\rho}{2}, 1 \right\}. \quad (2.5)$$

Thus, in the total-variation case,

$$Q_{\varphi, \rho}^{\text{WC}}(1 - \alpha; P_{\text{cal}}) = Q_{P_{\text{cal}}}(u_\rho^{\text{TV}}(\alpha)).$$

Equation (2.5) gives the main intuition needed later. The radius ρ increases the calibration level by $\rho/2$, with clipping at one. A larger radius protects against a larger class of possible test score distributions, but it also yields a larger threshold. If the clipped level equals one and the calibration score distribution is unbounded above, the resulting robust quantile is $+\infty$. Robust validation can therefore become conservative or even vacuous when the required protection against score-distribution shift is too large.

The discussion above is stated at the population level. Empirical finite-sample versions replace the unknown calibration score distribution by the empirical score distribution and require small finite-sample corrections to obtain exact coverage; see [Cauchois et al. \(2024, Theorem 1 and Corollaries 2.1–2.3\)](#) for the corresponding guarantees.

The worked population analysis in Chapter 3 uses the population version rather than the empirical finite-sample version. Thus P_{cal} , future score laws, and the corresponding robust quantiles are computed from the known data-generating mechanism. The robust-validation radius is an oracle quantity: for a single test score law it is $D_\varphi(P_{\text{test}} \| P_{\text{cal}})$, and for an admissible class of future score laws it is the supremum of this discrepancy over the

class. This oracle choice is not available in a real application. It is used here to isolate how the predictor affects the score-distribution discrepancy.

For a fixed predictor and score function, a larger radius leads to a more conservative quantile and wider prediction sets. If a different predictor induces calibration and future score laws that are closer to one another, the radius required for coverage can decrease. This observation motivates the predictor-side perspective taken next. Robust validation adjusts the calibration threshold for a given score-distribution discrepancy, whereas DRIG changes the predictor and hence the induced residual and score distributions before robust validation is applied.

2.4 Distributional Robustness via Invariant Gradients

2.4.1 Robust prediction across environments

DRIG is a predictor-side response to distribution shift. This matters for conformal prediction because the scores depend on the fitted predictor. In the regression setting considered in this thesis, the score is typically the absolute residual $s(x, y) = |y - \hat{f}(x)|$. Changing $\hat{f}$ therefore changes the residuals and the score distributions used for calibration.

This motivates robust prediction across environments. Suppose that data are observed under several environments, where each environment corresponds to a different data-generating regime. A predictor fitted only for optimal performance in one environment may perform poorly under future shifts. A robust prediction method instead uses the heterogeneity across observed environments to construct a predictor whose performance is more stable under plausible distributional changes.

Distributional Robustness via Invariant Gradients (DRIG), introduced by [Shen et al. \(2026\)](#), follows this idea. The method uses heterogeneous training data from several environments to learn predictors that are robust against a data-dependent class of unseen shifts. The robustness result used in this thesis is formulated in a linear structural equation model. In this model, the covariate-response vector in environment e is written as

$$Z^e := \begin{pmatrix} X^e \\ Y^e \end{pmatrix},$$

and satisfies

$$Z^e = BZ^e + \xi^e,$$

where B is a structural matrix and ξ^e is a noise vector. If $I - B$ is invertible, then $Z^e = (I - B)^{-1}\xi^e$. The environments and test interventions are represented by changes in the distribution of ξ^e , while B is kept fixed ([Shen et al., 2026](#), Section 2.1). Thus, in this setting, an intervention refers to a possible change in the SCM noise law.

In contrast to distributionally robust optimization (DRO) based on a pre-specified metric ball around the training distribution, DRIG uses the structure of the observed heterogeneity to determine the directions of the perturbations against which the predictor is protected. A tuning parameter γ controls the strength of this robustness.

For the present thesis, the important point is not the full generality of DRIG, but this predictor-side mechanism: a more stable residual distribution can lead to a more stable score distribution, which is the object robust validation controls.

2.4.2 The DRIG objective and a fixed-reference specialization

This subsection introduces the population DRIG objective from Shen et al. (2026) and then states a fixed-reference specialization used in the worked population analysis in Chapter 3.

Let $\mathcal{E}$ be a finite set of observed environments. For each $e \in \mathcal{E}$, let (X^e, Y^e) denote a covariate-response pair generated in environment e , where $X^e \in \mathbb{R}^p$ and $Y^e \in \mathbb{R}$. For a linear predictor $x \mapsto b^\top x$, define the population mean squared prediction risk in environment e by

$$R_e(b) := \mathbb{E}[(Y^e - b^\top X^e)^2], \quad b \in \mathbb{R}^p.$$

Let $\omega^e \geq 0$, $e \in \mathcal{E}$, be environment weights satisfying $\sum_{e \in \mathcal{E}} \omega^e = 1$.

Definition 2.6 (Population DRIG objective). *The population DRIG objective is*

$$L_\gamma(b) := \min_{e \in \mathcal{E}} R_e(b) + \gamma \sum_{e \in \mathcal{E}} \omega^e \left(R_e(b) - \min_{e' \in \mathcal{E}} R_{e'}(b) \right), \quad \gamma \geq 0. \quad (2.6)$$

A population DRIG coefficient is any minimizer

$$b_\gamma^{\text{opt}} \in \underset{b \in \mathbb{R}^p}{\operatorname{argmin}} L_\gamma(b).$$

This formulation follows the definitions from Shen et al. (2026, Equations (4)–(5)).

The first term in (2.6) is the smallest mean squared prediction risk across the observed environments. The second term penalizes deviations of the environment-specific risks from this smallest risk. Since $R_e(b) - \min_{e' \in \mathcal{E}} R_{e'}(b) \geq 0$ for all $e \in \mathcal{E}$, the penalty term is nonnegative. The parameter γ determines how strongly such cross-environment risk differences are penalized.

The objective is motivated by gradient invariance. A coefficient b satisfies the *gradient invariance condition* of Shen et al. (2026, Definition 1) if

$$\sum_{e \in \mathcal{E}} \omega^e \nabla_b R_e(b) = \nabla_b \min_{e \in \mathcal{E}} R_e(b),$$

at points where the right-hand side is differentiable. In words, the weighted average gradient of the environment-specific risks agrees with the gradient in an environment attaining the smallest risk. DRIG encourages this condition through the penalty term in (2.6). Small values of γ emphasize performance in the lowest-risk environment, whereas larger values of γ put more emphasis on cross-environment stability. In the settings analyzed by Shen et al. (2026), varying γ is used to trade off in-distribution prediction performance against stronger robustness to unseen interventions.

The minimum in (2.6) makes the DRIG objective piecewise quadratic in the linear-Gaussian settings considered in Chapter 3, because the environment attaining the smallest risk may depend on b . The worked population analysis therefore uses a fixed baseline environment instead. Let $0 \in \mathcal{E}$ denote this baseline environment.

Definition 2.7 (Fixed-reference DRIG objective). *The fixed-reference DRIG objective is*

$$L_\gamma^{\text{ref}}(b) := R_0(b) + \gamma \sum_{e \in \mathcal{E}} \omega^e (R_e(b) - R_0(b)), \quad \gamma \geq 0. \quad (2.7)$$

The objective in (2.7) measures deviations relative to the preselected baseline environment rather than relative to the smallest-risk environment.

The fixed-reference objective coincides with the paper-style objective at every b satisfying

$$R_0(b) = \min_{e \in \mathcal{E}} R_e(b).$$

In particular, if this equality holds for all $b \in \mathbb{R}^p$, then

$$L_\gamma^{\text{ref}}(b) = L_\gamma(b) \quad \text{for all } b \in \mathbb{R}^p \text{ and all } \gamma \geq 0.$$

This condition can hold in simple mean-shift constructions in which the baseline environment has uniformly smallest risk, but it is not automatic. In the mean-only Gaussian subclass of Section 3.6.1, for example, the baseline noise mean is zero and the covariance is fixed, so the baseline risk is uniformly smallest.

In general, the fixed-reference formulation should therefore be viewed as an analytically convenient specialization rather than as a replacement of the original DRIG definition. The convenience is mathematical: in the Gaussian population model used in Chapter 3, replacing the minimum-risk term by $R_0(b)$ yields a single quadratic objective and hence an explicit coefficient path. The limitation is that the resulting path need not coincide with the path obtained from the original DRIG objective when the smallest-risk environment changes with b .

The fixed-reference objective has a simple interpretation. For $\gamma = 0$, it reduces to the baseline-environment risk $R_0(b)$. Thus, its minimizer is the population ordinary least squares (OLS) coefficient in the baseline environment. For $\gamma = 1$, it becomes

$$L_1^{\text{ref}}(b) = R_0(b) + \sum_{e \in \mathcal{E}} \omega^e (R_e(b) - R_0(b)) = \sum_{e \in \mathcal{E}} \omega^e R_e(b),$$

because the weights sum to one. Hence, at $\gamma = 1$, minimizing L_1^{ref} is equivalent to minimizing the weighted pooled risk over the observed environments. For larger values of γ , the objective gives increasing weight to the risk contrasts between the observed environments and the baseline environment. If $R_0(b) \leq R_e(b)$ holds for a given b and environment e , then $R_e(b) - R_0(b)$ is the amount by which the risk in environment e exceeds the baseline risk.

This interpolation is central for the worked population analysis. For the fixed-reference objective, it provides a population path b_γ that starts at the baseline-environment OLS coefficient, passes through the weighted pooled OLS coefficient at $\gamma = 1$, and then moves toward predictors that place more weight on the environment heterogeneity encoded by the risk contrasts. Chapter 3 derives this path explicitly in a fixed- B Gaussian SCM.

2.4.3 Robustness interpretation and risk-level uncertainty sets

The main motivation for DRIG is not merely to penalize variation across observed risks, but to obtain robust prediction under unseen shifts. In the linear structural setting studied by Shen et al. (2026), test distributions are generated by the same structural matrix as the observed environments but with a possibly different noise vector v . The paper shows that the population DRIG solution solves a worst-case mean-squared-risk problem over a data-dependent class of such future noise distributions.

More precisely, let $S_e := \mathbb{E}[\xi^e(\xi^e)^\top]$ denote the second moment of the SCM noise vector in environment e .

Definition 2.8 (DRIG second-moment uncertainty set). *In the notation of this thesis, the second-moment uncertainty set corresponding to γ is*

$$\mathcal{C}_\gamma := \left\{ P_v : \mathbb{E}_{P_v}[vv^\top] \preceq S_0 + \gamma \sum_{e \in \mathcal{E}} \omega^e (S_e - S_0) \right\},$$

where v denotes a possible future SCM noise vector. Here $A \preceq B$ means that $B - A$ is positive semidefinite.

Under the linear SCM assumptions and the observational second-moment condition in Assumption 1 of Shen et al. (2026), DRIG has the following worst-case risk interpretation.

Theorem 2.9 (DRIG worst-case risk interpretation). *Under the assumptions of Shen et al. (2026, Theorem 3), the population DRIG coefficient solves the worst-case risk problem*

$$b_\gamma^{\text{opt}} \in \operatorname{argmin}_{b \in \mathbb{R}^p} \sup_{P_v \in \mathcal{C}_\gamma} \mathbb{E}_{P_v} \left[(Y^v - b^\top X^v)^2 \right].$$

This result is a risk-level robustness guarantee. It says that DRIG controls mean squared prediction risk over a class of future noise laws whose second moments are bounded by the observed environment heterogeneity. The parameter γ changes the size of this class, while the matrix $S_0 + \gamma \sum_{e \in \mathcal{E}} \omega^e (S_e - S_0)$ determines the directions of the protected noise interventions.

This risk-level guarantee is different from the score-level guarantee required by robust validation. Robust validation requires control of the distribution of the conformal score, for example through the total-variation distance between calibration and test score laws. A second-moment constraint on v controls the second moment of the residual, but it does not by itself control the full distribution of the absolute residual score. Section 3.5 uses this distinction to show that the full second-moment uncertainty set can be too large to give informative total-variation robust-validation radii.

Chapter 3 therefore does not use the DRIG robustness theorem as a black-box optimality result for the fixed-reference path. Instead, it uses the second-moment uncertainty-set construction as a risk-level object and asks what the explicitly defined fixed-reference coefficients b_γ imply for residual moments and score laws.

DRIG is also related to anchor regression, but this connection plays only a supporting role in this thesis. Anchor regression is a linear prediction method that uses anchor variables or environment labels to trade off prediction under the training distribution against stability under additive shift interventions (Rothenhäusler et al., 2021). Shen et al. (2026, Proposition 1) show that, under a discrete-anchor model with additive mean shifts and a reference environment with zero anchor value, the anchor-regression loss coincides with the DRIG loss. Thus, in mean-shift settings of this type, DRIG reduces to an anchor-regression-type objective.

The distinction becomes important beyond pure mean shifts. Anchor regression is designed for additive shift interventions, whereas DRIG is formulated for more general noise interventions. In particular, when the heterogeneous environments differ not only in means but also in second moments, DRIG can exploit additional structure that is not captured by anchor regression. This motivates the covariance-only setting and the setting based on the full DRIG second-moment uncertainty set studied in Chapters 3 and 4.

2.5 Combining robust prediction and robust calibration

The previous sections introduce two forms of robustness that act at different stages of the prediction pipeline. Robust validation modifies the calibration threshold for a fixed predictor, whereas DRIG changes the predictor itself. These two approaches interact because conformal prediction in regression uses scores that depend on the fitted predictor.

For the absolute residual score $s(x, y) = |y - \hat{f}(x)|$, changing $\hat{f}$ changes the calibration and test score distributions.

This motivates the central question of the thesis: *Can a robust predictor make robust calibration less conservative?* More concretely, Chapter 3 studies how the fixed-reference DRIG parameter γ changes the induced score laws, the total-variation robust-validation radius, and the resulting prediction interval width.

The answer cannot be derived from prediction risk alone. DRIG provides risk-level robustness, whereas robust validation requires score-level distributional control. Chapter 3 makes this distinction explicit by comparing the full second-moment uncertainty set with structured Gaussian subclasses for which risk-level control can be translated into score-level control.

Chapter 3

A Worked Population Analysis

Chapter 2 introduced robust validation as a calibration-side response to score-distribution shift and DRIG as a predictor-side response to environment heterogeneity. This chapter studies how these two approaches interact in a tractable population model.

For a linear predictor $x \mapsto b^\top x$, define the absolute residual score in environment e by

$$T_b^e := |Y^e - b^\top X^e|.$$

Changing b changes the residuals and therefore the score distributions. The central question in this chapter is how the path b_γ changes the robust-validation radius and the resulting interval width.

The analysis is carried out at the population level. All risks, residual laws, score laws, radii, coverages, and widths in this chapter are population quantities. This removes finite-sample estimation effects and isolates the mechanism linking predictor-side robustness to calibration-side robustness.

Building on the fixed- B structural equation framework introduced in Chapter 2, we specialize to Gaussian noise laws whose means and covariances may vary across environments. This setting is broad enough to represent several forms of environment heterogeneity, but remains analytically tractable: risks are quadratic in the predictor, residuals are Gaussian, and absolute residual scores have folded-normal distributions.

The chapter proceeds as follows. Section 3.1 defines the fixed- B Gaussian SCM and introduces the moment notation used throughout the chapter. Section 3.2 derives the coefficient path used below. Section 3.3 computes the residual and score distributions induced by this path. Section 3.4 defines the corresponding conformal and robust-validation quantities. Section 3.5 compares this score-distribution analysis with the risk-level second-moment uncertainty set. Section 3.6 identifies structured Gaussian subclasses for which risk-level control implies score-level control.

3.1 A fixed- B Gaussian SCM

Let $\mathcal{E}_{\text{obs}} = \{0, 1, \dots, m\}$ be a finite set of observed environments, where $0 \in \mathcal{E}_{\text{obs}}$ denotes the baseline (or reference) environment. Let ω^e , $e \in \mathcal{E}_{\text{obs}}$, be convex environment weights, i.e. $\omega^e \geq 0$ and $\sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e = 1$. The observed environments are used to define the DRIG objective, the calibration score law, and the second-moment matrices used in Section 3.5.

For each $e \in \mathcal{E}_{\text{obs}}$, we write (X^e, Y^e) for a generic population draw from that environment. Here $X^e \in \mathbb{R}^p$ is the covariate vector and $Y^e \in \mathbb{R}$ is the response. Define

$$Z^e := \begin{pmatrix} X^e \\ Y^e \end{pmatrix} \in \mathbb{R}^{p+1}.$$

We assume that Z^e follows a linear SCM with a fixed structural matrix $B \in \mathbb{R}^{(p+1) \times (p+1)}$:

$$Z^e = BZ^e + \xi^e.$$

This is the structural representation used in the linear DRIG framework (Shen et al., 2026), as discussed in Sections 2.2.1 and 2.4.3.

Assume that $I - B$ is invertible and set $K := (I - B)^{-1}$. Then $Z^e = K\xi^e$. The matrix B is kept fixed across environments, so environment heterogeneity enters through the distribution of the noise vector ξ^e .

For each $e \in \mathcal{E}_{\text{obs}}$, assume that

$$\xi^e \sim N(\delta^e, \Omega_e),$$

where $\delta^e \in \mathbb{R}^{p+1}$ and $\Omega_e \in \mathbb{R}^{(p+1) \times (p+1)}$ satisfies $\Omega_e \succ 0$. Hence environments may differ in their noise means and covariance matrices. Both parts are relevant for the risk calculations below, because the second moment of ξ^e is $\Omega_e + \delta^e(\delta^e)^\top$. The positive definiteness assumption excludes degenerate Gaussian laws and ensures that the induced residual score laws have densities.

As in Chapter 2, the structural equations are used as a tractable model for shifts in the joint distribution. A causal interpretation of B would require additional assumptions. The calculations below use the structural equations only as a tractable population model.

The central moment object in the following risk calculations is the second-moment matrix of Z^e ,

$$\Lambda_e := \mathbb{E}[Z^e(Z^e)^\top].$$

Since $Z^e = K\xi^e$, this matrix is

$$\Lambda_e = K \left(\Omega_e + \delta^e(\delta^e)^\top \right) K^\top.$$

We partition Λ_e according to the covariate and response coordinates:

$$\Lambda_e = \begin{pmatrix} \Lambda_{XX}^e & \Lambda_{XY}^e \\ (\Lambda_{XY}^e)^\top & \Lambda_{YY}^e \end{pmatrix},$$

where $\Lambda_{XX}^e = \mathbb{E}[X^e(X^e)^\top]$, $\Lambda_{XY}^e = \mathbb{E}[X^e Y^e]$, and $\Lambda_{YY}^e = \mathbb{E}[(Y^e)^2]$.

The matrix Λ_e will be used to express mean squared prediction risk as a quadratic function of b . For robust validation, however, the full residual distribution is needed. Section 3.3 obtains this distribution by writing the residual as a linear function of the Gaussian noise vector ξ^e .

3.2 The fixed-reference DRIG path

This section derives the predictor path used throughout the rest of the chapter. The goal is to obtain an explicit expression for the fixed-reference DRIG coefficient b_γ in the Gaussian SCM of Section 3.1. Once this path is known, the later sections can study how changing γ changes residual laws, score laws, robust-validation radii, and interval widths.

The derivation has three steps. First, the squared prediction risk is written as a quadratic function of b . Second, the baseline and pooled OLS coefficients are identified as two reference points. Third, the fixed-reference DRIG objective is expanded as a quadratic objective whose minimizer is explicit. The only output needed later is the coefficient b_γ and the corresponding residual direction w_γ .

3.2.1 Residual and risk representation

The first step is to express residuals and risks in terms of the moment matrices from Section 3.1. For a linear predictor $x \mapsto b^\top x$, define

$$\ell_b := \begin{pmatrix} -b \\ 1 \end{pmatrix} \in \mathbb{R}^{p+1}.$$

The residual in environment e is $U^e(b) := Y^e - b^\top X^e = \ell_b^\top Z^e$. Since $Z^e = K\xi^e$, this can be written as

$$U^e(b) = \ell_b^\top K\xi^e = w_b^\top \xi^e, \quad w_b := K^\top \ell_b.$$

Thus, for a fixed predictor b , the residual is a one-dimensional projection of the SCM noise vector. This projection is the link between the predictor and the residual distribution studied below.

Proposition 3.1 (Quadratic risk representation). *Under the fixed-B Gaussian SCM, for every $e \in \mathcal{E}_{\text{obs}}$ and every $b \in \mathbb{R}^p$,*

$$R_e(b) := \mathbb{E} \left[(Y^e - b^\top X^e)^2 \right] = \ell_b^\top \Lambda_e \ell_b.$$

Equivalently,

$$R_e(b) = b^\top \Lambda_{XX}^e b - 2b^\top \Lambda_{XY}^e + \Lambda_{YY}^e.$$

The proof is given in Appendix A.1.

3.2.2 Reference and pooled OLS

The next step is to identify two benchmark coefficients that serve as reference points for the fixed-reference path. At $\gamma = 0$, the path will recover the OLS coefficient in the baseline environment. At $\gamma = 1$, it will recover the OLS coefficient for the weighted observed risk.

First consider the baseline environment. Since $I - B$ is invertible and $\Omega_e \succ 0$, the matrix Λ_{XX}^e is positive definite for every $e \in \mathcal{E}_{\text{obs}}$. In particular, $\Lambda_{XX}^0 \succ 0$. Hence the baseline risk $R_0(b)$ has the unique minimizer

$$b_{\text{ref}}^{\text{OLS}} = (\Lambda_{XX}^0)^{-1} \Lambda_{XY}^0.$$

For the observed environments, define the weighted moment summaries

$$\bar{\Lambda}_{XX, \text{obs}} := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e \Lambda_{XX}^e, \quad \bar{\Lambda}_{XY, \text{obs}} := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e \Lambda_{XY}^e, \quad \bar{\Lambda}_{YY, \text{obs}} := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e \Lambda_{YY}^e.$$

The weighted observed risk is $\bar{R}_{\text{obs}}(b) := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e R_e(b)$. Using Proposition 3.1, it can be written as

$$\bar{R}_{\text{obs}}(b) = b^\top \bar{\Lambda}_{XX,\text{obs}} b - 2b^\top \bar{\Lambda}_{XY,\text{obs}} + \bar{\Lambda}_{YY,\text{obs}}.$$

Since $\bar{\Lambda}_{XX,\text{obs}}$ is a nonnegative weighted sum of positive definite matrices with weights summing to one, it is positive definite. Hence the weighted observed risk has the unique minimizer

$$b_{\text{pld}}^{\text{OLS}} = \bar{\Lambda}_{XX,\text{obs}}^{-1} \bar{\Lambda}_{XY,\text{obs}}.$$

The two coefficients $b_{\text{ref}}^{\text{OLS}}$ and $b_{\text{pld}}^{\text{OLS}}$ serve as benchmarks. They will appear as the $\gamma = 0$ and $\gamma = 1$ cases of the path below.

3.2.3 Fixed-reference objective and coefficient path

The final step is to interpolate between these two risk objectives through the fixed-reference DRIG objective. For $\gamma \geq 0$, define

$$L_\gamma^{\text{ref}}(b) := R_0(b) + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (R_e(b) - R_0(b)).$$

Equivalently,

$$L_\gamma^{\text{ref}}(b) = (1 - \gamma)R_0(b) + \gamma \bar{R}_{\text{obs}}(b).$$

This is the specialization discussed in Section 2.4.2. In the present Gaussian model, it is used because it yields an explicit quadratic path. If $R_0(b) = \min_{e \in \mathcal{E}_{\text{obs}}} R_e(b)$ for all $b \in \mathbb{R}^p$, it agrees with the corresponding minimum-risk formulation. Otherwise, it should be read as an analytical specialization.

The terms $R_e(b) - R_0(b)$ are risk contrasts relative to the baseline and need not be non-negative for all b . This does not prevent the path from being well-defined. The relevant condition is strict convexity of the resulting quadratic objective.

Define

$$A_\gamma := \Lambda_{XX}^0 + \gamma (\bar{\Lambda}_{XX,\text{obs}} - \Lambda_{XX}^0), \quad h_\gamma := \Lambda_{XY}^0 + \gamma (\bar{\Lambda}_{XY,\text{obs}} - \Lambda_{XY}^0),$$

and

$$c_\gamma := \Lambda_{YY}^0 + \gamma (\bar{\Lambda}_{YY,\text{obs}} - \Lambda_{YY}^0).$$

With this notation, the fixed-reference objective has the following explicit minimizer.

Proposition 3.2 (Fixed-reference DRIG coefficient path). *For every $\gamma \geq 0$, L_γ^{ref} has the form*

$$L_\gamma^{\text{ref}}(b) = b^\top A_\gamma b - 2b^\top h_\gamma + c_\gamma.$$

If $A_\gamma \succ 0$, then L_γ^{ref} has the unique minimizer

$$b_\gamma = A_\gamma^{-1} h_\gamma.$$

In particular,

$$b_0 = b_{\text{ref}}^{\text{OLS}}, \quad b_1 = b_{\text{pld}}^{\text{OLS}}.$$

The proof is given in Appendix A.2. It expands L_γ^{ref} using Proposition 3.1 and solves the resulting quadratic optimization problem.

Let $\Gamma := \{\gamma \geq 0 : A_\gamma \succ 0\}$. For $\gamma \in \Gamma$, the objective L_γ^{ref} is strictly convex because its Hessian with respect to b is $2A_\gamma$. We restrict the path to Γ so that the minimizer is unique and the predictor path is well-defined.

The identity $A_\gamma = (1 - \gamma)\Lambda_{XX}^0 + \gamma\bar{\Lambda}_{XX,\text{obs}}$ holds for all $\gamma \geq 0$. For $\gamma \in [0, 1]$, this is a convex combination of two positive definite matrices. Hence $A_\gamma \succ 0$ for every $\gamma \in [0, 1]$, and therefore $[0, 1] \subset \Gamma$. For $\gamma > 1$, positive definiteness is not automatic and remains a well-posedness condition.

Throughout the remainder of the thesis, b_γ denotes the coefficient from Proposition 3.2 whenever $\gamma \in \Gamma$. We also write

$$w_\gamma := w_{b_\gamma} = K^\top \ell_{b_\gamma}$$

for the corresponding residual direction.

The coefficient b_γ links the quadratic objective to the score distributions used for calibration. Changing γ changes b_γ , and hence changes w_γ . The next section computes the induced residual and score distributions.

3.3 Residual and score distributions along the DRIG path

The path b_γ determines the predictor, while robust validation is applied to absolute residual scores. We therefore compute the score distributions induced by b_γ . In the fixed- B Gaussian SCM, these distributions are explicit because each residual is a linear function of a Gaussian noise vector.

For $\gamma \in \Gamma$ and $e \in \mathcal{E}_{\text{obs}}$, define

$$U_\gamma^e := Y^e - b_\gamma^\top X^e.$$

Using the notation from Section 3.2.1,

$$U_\gamma^e = w_\gamma^\top \xi^e.$$

Thus, for a fixed value of γ , the residual is a one-dimensional projection of the Gaussian SCM noise vector.

Lemma 3.3 (Residual law along the fixed-reference path). *For every $\gamma \in \Gamma$ and every $e \in \mathcal{E}_{\text{obs}}$,*

$$U_\gamma^e = w_\gamma^\top \xi^e \sim N(\mu_{e,\gamma}, \sigma_{e,\gamma}^2), \quad w_\gamma = K^\top \begin{pmatrix} -b_\gamma \\ 1 \end{pmatrix},$$

where

$$\mu_{e,\gamma} = w_\gamma^\top \delta^e, \quad \sigma_{e,\gamma}^2 = w_\gamma^\top \Omega_e w_\gamma.$$

For a fixed environment e , the quantities δ^e and Ω_e do not depend on γ . The dependence on γ enters through the predictor b_γ , hence through the projection direction w_γ . Thus the DRIG path changes the residual distribution by changing the projected mean $w_\gamma^\top \delta^e$ and projected variance $w_\gamma^\top \Omega_e w_\gamma$.

The proof follows directly from the fact that linear transformations of Gaussian random vectors are Gaussian. Lemma 3.3 shows the mechanism by which the DRIG path affects robust validation: changing γ can change both the projected residual mean and the projected residual variance, and hence the distribution of the absolute residual score.

The absolute residual score in environment e is

$$T_\gamma^e := |U_\gamma^e|.$$

Since U_γ^e is Gaussian, T_γ^e has a folded-normal distribution. For $x \geq 0$, its distribution function is

$$F_{e,\gamma}(x) := \mathbb{P}(T_\gamma^e \leq x) = \Phi\left(\frac{x - \mu_{e,\gamma}}{\sigma_{e,\gamma}}\right) - \Phi\left(\frac{-x - \mu_{e,\gamma}}{\sigma_{e,\gamma}}\right),$$

where Φ denotes the standard normal distribution function. Its density on $[0, \infty)$ is

$$f_{e,\gamma}(x) = \frac{1}{\sigma_{e,\gamma}} \phi\left(\frac{x - \mu_{e,\gamma}}{\sigma_{e,\gamma}}\right) + \frac{1}{\sigma_{e,\gamma}} \phi\left(\frac{x + \mu_{e,\gamma}}{\sigma_{e,\gamma}}\right),$$

where ϕ denotes the standard normal density.

The calibration score distribution is the weighted mixture of the observed environment score distributions. We define its distribution function by

$$F_{\text{cal},\gamma}(x) := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e F_{e,\gamma}(x), \quad x \geq 0.$$

Since this is a finite mixture of distributions with densities, its density is

$$f_{\text{cal},\gamma}(x) := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e f_{e,\gamma}(x).$$

We write $T_{\text{cal},\gamma}$ for a random variable with distribution function $F_{\text{cal},\gamma}$.

To define the robust-validation radius, we also need the score law under a structured test shift. A structured test shift t is modeled in the same fixed- B structural representation, but with a new Gaussian noise vector $v^t \sim N(\delta^t, \Omega_t)$, where $\Omega_t \succ 0$. This test shift is not used to construct the predictor path or the calibration score law. The induced residual under b_γ is

$$U_\gamma^t := w_\gamma^\top v^t.$$

By the same argument as in Lemma 3.3,

$$U_\gamma^t \sim N(\mu_{t,\gamma}, \sigma_{t,\gamma}^2), \quad \mu_{t,\gamma} = w_\gamma^\top \delta^t, \quad \sigma_{t,\gamma}^2 = w_\gamma^\top \Omega_t w_\gamma.$$

The corresponding absolute residual score has distribution function $F_{t,\gamma}$ and density $f_{t,\gamma}$, given by the same folded-normal formulas as above.

The robust-validation radius studied below compares future score distributions induced by admissible noise distributions with the calibration score distribution $F_{\text{cal},\gamma}$.

3.4 Population conformal prediction and robust validation

The previous section derived the calibration and test score laws induced by b_γ . This section turns these laws into conformal thresholds, coverages, and widths.

3.4.1 Standard conformal intervals

Fix a target miscoverage level $\alpha \in (0, 1)$. The population analogue of the standard conformal threshold is the $(1 - \alpha)$ -quantile of the calibration score law:

$$q_\gamma^{\text{std}}(\alpha) := \inf\{x \geq 0 : F_{\text{cal},\gamma}(x) \geq 1 - \alpha\}.$$

The corresponding prediction interval at covariate value $x \in \mathbb{R}^p$ is

$$C_\gamma^{\text{std}}(x) := \left[b_\gamma^\top x - q_\gamma^{\text{std}}(\alpha), b_\gamma^\top x + q_\gamma^{\text{std}}(\alpha) \right].$$

If the test score law equals the calibration score law, the interval has population coverage $1 - \alpha$. Under distribution shift, the test score law may differ from $F_{\text{cal},\gamma}$. For a structured test shift t , the population coverage of the interval is

$$\text{Cov}_{t,\gamma}^{\text{std}} := F_{t,\gamma}(q_\gamma^{\text{std}}(\alpha)).$$

Thus, standard conformal calibration undercovers for this shift whenever $F_{t,\gamma}(q_\gamma^{\text{std}}(\alpha)) < 1 - \alpha$.

The standard interval width is $\text{Width}_\gamma^{\text{std}} := 2q_\gamma^{\text{std}}(\alpha)$. This width depends on γ because b_γ changes the calibration score law $F_{\text{cal},\gamma}$.

3.4.2 Robust-validation intervals

This subsection uses the total-variation version of robust validation introduced in Section 2.3. Unless stated otherwise, robust validation in the remainder of the population analysis refers to this version.

First consider a single Gaussian test shift t as in Section 3.3. Its radius is

$$\rho_t(\gamma) := \int_0^\infty |f_{t,\gamma}(x) - f_{\text{cal},\gamma}(x)| dx.$$

Since the score laws have densities, this quantity equals twice the total variation distance between the corresponding score distributions:

$$\rho_t(\gamma) = 2 \left\| \mathcal{L}(T_\gamma^t) - \mathcal{L}(T_{\text{cal},\gamma}) \right\|_{\text{TV}}.$$

The next sections compare these score-level quantities with DRIG-type uncertainty sets for future SCM noise laws. For this purpose, let $\mathcal{A}_\gamma$ be a class of possible future laws for the SCM noise v . For $P_v \in \mathcal{A}_\gamma$, with $v \sim P_v$, let

$$P_{v,\gamma}^T := \mathcal{L}(|w_\gamma^\top v|)$$

denote the induced score law. Define $\rho_v(\gamma) := 2 \left\| P_{v,\gamma}^T - \mathcal{L}(T_{\text{cal},\gamma}) \right\|_{\text{TV}}$. The worst-case radius over $\mathcal{A}_\gamma$ is

$$\rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma) := \sup_{P_v \in \mathcal{A}_\gamma} \rho_v(\gamma).$$

When the admissible class is clear from context, we write $\rho_{\text{wc}}(\gamma)$.

Using the total-variation correction from Section 2.3, the robust-validation quantile level associated with $\mathcal{A}_\gamma$ is

$$u_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) := \min \left\{ 1 - \alpha + \frac{\rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma)}{2}, 1 \right\}.$$

The corresponding robust threshold is

$$q_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) := \inf \left\{ x \geq 0 : F_{\text{cal}, \gamma}(x) \geq u_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) \right\},$$

with the convention $q_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) = +\infty$ if $u_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) = 1$. The robust-validation interval is

$$C_\gamma^{\text{RV}}(x; \mathcal{A}_\gamma) := \left[b_\gamma^\top x - q_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma), b_\gamma^\top x + q_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma) \right].$$

Proposition 3.4 (Population robust-validation guarantee). *Let $\gamma \in \Gamma$ and let $\mathcal{A}_\gamma$ be a class of possible future laws for the SCM noise v . Then the robust-validation interval $C_\gamma^{\text{RV}}(\cdot; \mathcal{A}_\gamma)$ satisfies*

$$\mathbb{P}_{P_v} \left(Y \in C_\gamma^{\text{RV}}(X; \mathcal{A}_\gamma) \right) \geq 1 - \alpha$$

for every $P_v \in \mathcal{A}_\gamma$. Here $\mathbb{P}_{P_v}$ denotes probability under the fixed-B SCM when the future noise vector satisfies $v \sim P_v$.

Indeed, for every $P_v \in \mathcal{A}_\gamma$, the definition of $\rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma)$ gives

$$2 \left\| P_{v, \gamma}^T - \mathcal{L}(T_{\text{cal}, \gamma}) \right\|_{\text{TV}} \leq \rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma).$$

The robust-validation guarantee from Section 2.3 therefore applies.

The corresponding interval width is

$$\text{Width}_\gamma^{\text{RV}}(\mathcal{A}_\gamma) := 2q_\gamma^{\text{RV}}(\alpha; \mathcal{A}_\gamma).$$

This width depends on γ in two ways. First, γ changes the calibration score law $F_{\text{cal}, \gamma}$. Second, γ changes the worst-case discrepancy $\rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma)$.

A width-optimal value of the DRIG tuning parameter for the indexed admissible classes $(\mathcal{A}_\gamma)_{\gamma \in \Gamma}$ is any

$$\gamma_{\text{width}}^* \in \underset{\gamma \in \Gamma}{\text{argmin}} \text{Width}_\gamma^{\text{RV}}(\mathcal{A}_\gamma),$$

whenever such a minimizer exists.

The densities $f_{e, \gamma}$ are explicit folded-normal densities, but the radius generally contains the absolute difference between folded-normal densities or mixtures. In the numerical study, these radii are evaluated by one-dimensional numerical integration or by the one-dimensional reductions implied by the structured subclasses in Section 3.6.

The construction above gives the population objects used in the numerical analysis:

$$\gamma \mapsto b_\gamma \mapsto F_{\text{cal}, \gamma}, P_{v, \gamma}^T \mapsto \rho_{\text{wc}}(\gamma; \mathcal{A}_\gamma) \mapsto \text{Width}_\gamma^{\text{RV}}(\mathcal{A}_\gamma).$$

The next section compares this score-distribution analysis with the risk-level uncertainty set associated with DRIG.

3.5 Risk-level robustness and score-level limitations

The DRIG robustness interpretation is formulated at the level of mean squared prediction risk. In the notation of the fixed- B SCM, let

$$S_e := \mathbb{E}[\xi^e(\xi^e)^\top] = \Omega_e + \delta^e(\delta^e)^\top.$$

Following the second-moment uncertainty-set construction used in [Shen et al. \(2026\)](#), define

$$S_\gamma := S_0 + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (S_e - S_0).$$

For $\gamma \in [0, 1]$, this matrix is positive semidefinite because it is a convex combination of S_0 and $\sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e S_e$. For $\gamma > 1$, positive semidefiniteness is not automatic. In what follows, the set below is used only for values of γ for which $S_\gamma \succeq 0$.

Definition 3.5 (Full second-moment uncertainty set). *For every γ such that $S_\gamma \succeq 0$, define*

$$\mathcal{C}_\gamma := \{P_v : \mathbb{E}[vv^\top] \preceq S_\gamma\},$$

where v denotes a possible future SCM noise vector.

For the predictor b_γ , the corresponding residual is $U_\gamma(v) = w_\gamma^\top v$. Hence every $P_v \in \mathcal{C}_\gamma$ satisfies

$$\mathbb{E}[U_\gamma(v)^2] = w_\gamma^\top \mathbb{E}[vv^\top] w_\gamma \leq w_\gamma^\top S_\gamma w_\gamma.$$

Thus, the second-moment uncertainty set gives a risk-level residual moment bound for b_γ .

Robust validation requires a different type of control. In the total-variation case, one needs the induced score law $\mathcal{L}(|w_\gamma^\top v|)$ to remain close to the calibration score law. The next result shows that a second-moment constraint alone is not enough for this purpose.

The supremum in the following result is taken over all probability laws P_v satisfying the second-moment constraint. No Gaussian assumption is imposed.

Proposition 3.6 (Second-moment uncertainty does not control the TV radius). *Let $S \succeq 0$ and let $w \in \mathbb{R}^{p+1}$ satisfy*

$$C := w^\top S w > 0.$$

Let P_{cal}^T be a continuous calibration score distribution on $[0, \infty)$. Then

$$\sup_{P_v : \mathbb{E}[vv^\top] \preceq S} 2 \left\| \mathcal{L}(|w^\top v|) - P_{\text{cal}}^T \right\|_{\text{TV}} = 2.$$

The proof is given in [Appendix A.3](#). It constructs a probability law that satisfies the constraint $\mathbb{E}[vv^\top] \preceq S$ by putting all mass at a vector q with $qq^\top \preceq S$ and $|w^\top q| = \sqrt{C}$. The induced score law is then a point mass at $\sqrt{C}$. Since a continuous calibration score distribution assigns no mass to this point, the total-variation distance is maximal.

Applied with $S = S_\gamma$, $w = w_\gamma$, and $P_{\text{cal}}^T = \mathcal{L}(T_{\text{cal}, \gamma})$, [Proposition 3.6](#) shows that the worst-case total-variation radius over the full second-moment set is maximal whenever $M_\gamma = w_\gamma^\top S_\gamma w_\gamma > 0$. The calibration score law is continuous here because it is a finite mixture of folded-normal distributions.

For the correction used in Section 3.4.2,

$$u_{\gamma}^{\text{RV}}(\alpha) = \min \left\{ 1 - \alpha + \frac{\rho}{2}, 1 \right\},$$

the robust quantile level reaches 1 whenever $\rho \geq 2\alpha$. In the full second-moment case, the worst-case radius is $\rho = 2$, so the robust quantile level is clipped at 1 for every $\alpha \in (0, 1)$. Thus, the full second-moment set is too broad to give informative robust-validation widths.

This does not mean that second-moment information is useless for prediction intervals. It means that such information does not, by itself, control the total-variation distance between score laws.

The same second-moment bound can still be useful under an additional Gaussian restriction. Let

$$M_{\gamma} := w_{\gamma}^{\top} S_{\gamma} w_{\gamma}.$$

Assume that $M_{\gamma} > 0$ and that the future SCM noise is Gaussian,

$$v \sim N(\mu, \Sigma),$$

with $\mathbb{E}[vv^{\top}] = \mu\mu^{\top} + \Sigma \preceq S_{\gamma}$. Then

$$U_{\gamma}(v) = w_{\gamma}^{\top} v \sim N(m_{\gamma}, \sigma_{\gamma}^2),$$

where $m_{\gamma} = w_{\gamma}^{\top} \mu$, $\sigma_{\gamma}^2 = w_{\gamma}^{\top} \Sigma w_{\gamma}$. Moreover,

$$m_{\gamma}^2 + \sigma_{\gamma}^2 = w_{\gamma}^{\top} (\mu\mu^{\top} + \Sigma) w_{\gamma} \leq M_{\gamma}.$$

Proposition 3.7 (Gaussian second-moment interval). *Let $Y \sim N(m, \sigma^2)$, and assume that*

$$m^2 + \sigma^2 \leq M$$

for some $M > 0$. Then, for every $\alpha \in (0, 1)$,

$$\mathbb{P} \left(|Y| \leq \sqrt{M} \sqrt{1 + z_{1-\alpha/2}^2} \right) \geq 1 - \alpha,$$

where $z_{1-\alpha/2} := \Phi^{-1}(1 - \frac{\alpha}{2})$.

The proof is given in Appendix A.4. If $M_{\gamma} > 0$, applying Proposition 3.7 to $Y = U_{\gamma}(v)$ gives

$$\mathbb{P} \left(|U_{\gamma}(v)| \leq \sqrt{M_{\gamma}} \sqrt{1 + z_{1-\alpha/2}^2} \right) \geq 1 - \alpha$$

for every Gaussian future noise law satisfying $\mu\mu^{\top} + \Sigma \preceq S_{\gamma}$. Equivalently, the interval

$$\left[b_{\gamma}^{\top} x - \sqrt{M_{\gamma}} \sqrt{1 + z_{1-\alpha/2}^2}, b_{\gamma}^{\top} x + \sqrt{M_{\gamma}} \sqrt{1 + z_{1-\alpha/2}^2} \right]$$

has coverage at least $1 - \alpha$ for every such Gaussian future noise law.

This interval is not a robust-validation interval. It does not use the calibration score law $F_{\text{cal}, \gamma}$, and it does not require a bound on the distance between calibration and test score laws. Instead, it is a direct Gaussian consequence of the residual second-moment bound. It is therefore useful as a comparison point for the setting based on the full DRIG second-moment uncertainty set, where the set is too broad for robust validation but still gives moment-level information.

The conclusion is that risk-level robustness and score-level robustness are related but distinct. The second-moment uncertainty set controls residual second moments, which is enough for risk statements and for Gaussian moment-based intervals. It does not by itself control the full distribution of the absolute residual score. To obtain informative robust-validation guarantees, one needs additional structure that links the DRIG penalty to the score distribution. The next section identifies two such structured Gaussian subclasses.

3.6 Structured Gaussian subclasses

Section 3.5 shows that the full second-moment uncertainty set associated with DRIG is too broad for an informative worst-case analysis based on total variation. This does not mean that robust validation cannot be informative under DRIG-type perturbations. It means that one needs additional structure connecting the risk-level penalty to the induced score distributions.

This section studies two such structured Gaussian subclasses. In the first, only the noise mean varies while the covariance is fixed. In the second, only the noise covariance varies while the mean is fixed. These restrictions give sufficient conditions under which the DRIG penalty controls a score-relevant projected quantity: projected residual means in the mean-only case and projected residual variances in the covariance-only case.

The proofs of both subclass results use the same deterministic quadratic penalty argument.

Lemma 3.8 (Quadratic penalty rate). *Let $L_\gamma(b) = a(b) + \gamma g(b)$, $\gamma \geq 0$, where $a : \mathbb{R}^p \rightarrow \mathbb{R}$ is a strictly convex quadratic function satisfying $a(b) \rightarrow \infty$ whenever $\|b\| \rightarrow \infty$, and $g : \mathbb{R}^p \rightarrow [0, \infty)$ is a nonnegative quadratic function. Assume that*

$$\mathcal{N} := \{b \in \mathbb{R}^p : g(b) = 0\} \neq \emptyset, \quad b_\infty := \operatorname{argmin}_{b \in \mathcal{N}} a(b),$$

and, for each $\gamma \geq 0$, let

$$b_\gamma := \operatorname{argmin}_{b \in \mathbb{R}^p} \{a(b) + \gamma g(b)\}.$$

Then, as $\gamma \rightarrow \infty$,

$$b_\gamma \rightarrow b_\infty, \quad \gamma g(b_\gamma) \rightarrow 0.$$

The proof is given in Appendix A.5. The rate $\gamma g(b_\gamma) \rightarrow 0$ is the key point for the two applications below. The admissible test classes grow with γ , so it is not enough that the projected penalty $g(b_\gamma)$ tends to zero. It has to vanish fast enough to compensate for this growth.

3.6.1 Mean-only Gaussian shifts

Assume first that the observed environments differ only through the mean of the SCM noise:

$$\xi^e \sim N(\delta^e, \Omega), \quad e \in \mathcal{E}_{\text{obs}},$$

where $\Omega \succ 0$ is common to all observed environments and $\delta^0 = 0$. Define

$$M_{\text{obs}} := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e \delta^e (\delta^e)^\top.$$

Under these assumptions, the second-moment constraint for a future Gaussian law $N(\delta, \Omega)$ reduces to $\delta\delta^\top \preceq \gamma M_{\text{obs}}$. This motivates the mean-only Gaussian class

$$\mathcal{C}_\gamma^{\text{mean}, G} := \left\{ N(\delta, \Omega) : \delta\delta^\top \preceq \gamma M_{\text{obs}} \right\},$$

where the superscript G indicates that the class is restricted to Gaussian laws. The class keeps the covariance fixed and restricts admissible test mean shifts to directions and magnitudes controlled by the observed mean shifts.

Proposition 3.9 (Mean-only Gaussian shifts). *Define the mean-invariance set*

$$\mathcal{N}_{\text{mean}} := \left\{ b \in \mathbb{R}^p : w_b^\top M_{\text{obs}} w_b = 0 \right\}. \quad (3.1)$$

Assume that $\mathcal{N}_{\text{mean}}$ is nonempty, and let

$$b_\infty^{\text{mean}} := \underset{b \in \mathcal{N}_{\text{mean}}}{\operatorname{argmin}} R_0(b) = \underset{b \in \mathcal{N}_{\text{mean}}}{\operatorname{argmin}} w_b^\top \Omega w_b.$$

Then the worst-case total-variation radius over the mean-only Gaussian class satisfies

$$\rho_{\text{wc}}(\gamma; \mathcal{C}_\gamma^{\text{mean}, G}) = \sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - \mathcal{L}(T_{\text{cal}, \gamma}) \right\|_{\text{TV}} \longrightarrow 0 \quad \text{as } \gamma \rightarrow \infty,$$

where $v \sim N(\delta, \Omega)$.

The proof is given in Appendix A.6. Under the condition $w_b^\top M_{\text{obs}} w_b = 0$, Lemma 3.8 drives the worst-case projected test mean to zero along the path b_γ . Since the covariance is fixed, this aligns the folded-normal score distributions in total variation.

Remark (Interpretation of the mean-invariance condition). The condition $w_b^\top M_{\text{obs}} w_b = 0$ is a residual-level invariance condition. Since

$$w_b^\top M_{\text{obs}} w_b = \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (w_b^\top \delta^e)^2,$$

it is equivalent to $w_b^\top \delta^e = 0$ for all observed environments with positive weight. In the centered mean-only setting, where $\delta^0 = 0$, this means that the residual mean is zero in all observed environments for this predictor. Thus, the assumption is not automatic for arbitrary Gaussian environments. It requires that there exists a linear predictor whose residual is invariant to the observed mean-shift directions.

3.6.2 Covariance-only Gaussian shifts

Next assume that the observed environments have a common noise mean but different covariance matrices:

$$\xi^e \sim N(\delta, \Omega_e), \quad \Omega_e = \Omega_0 + H_e, \quad e \in \mathcal{E}_{\text{obs}},$$

where $\Omega_0 \succ 0$ is the baseline covariance matrix and $H_0 = 0$. We impose the structural condition

$$H_e \succeq 0 \quad \text{for all } e \in \mathcal{E}_{\text{obs}}.$$

This condition rules out cancellations between positive and negative covariance perturbations. It ensures that if the average projected covariance perturbation is zero, then each observed projected perturbation is zero.

Define

$$\Delta_{\text{obs}} := \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e H_e.$$

For a future Gaussian law $N(\delta, \Omega_0 + H)$, the second-moment constraint gives $H \preceq \gamma \Delta_{\text{obs}}$. We additionally restrict to positive semidefinite covariance perturbations and define

$$\mathcal{C}_\gamma^{\text{cov}, G} := \{N(\delta, \Omega_0 + H) : 0 \preceq H \preceq \gamma \Delta_{\text{obs}}\}.$$

The superscript G again indicates that the class is restricted to Gaussian laws. The class keeps the noise mean fixed and restricts admissible covariance perturbations to positive semidefinite perturbations controlled by the observed covariance heterogeneity.

Proposition 3.10 (Covariance-only Gaussian shifts). *Assume that*

$$\mathcal{N}_{\text{cov}} := \left\{b \in \mathbb{R}^p : w_b^\top \Delta_{\text{obs}} w_b = 0\right\}$$

is nonempty, and let

$$b_\infty^{\text{cov}} := \underset{b \in \mathcal{N}_{\text{cov}}}{\operatorname{argmin}} R_0(b) = \underset{b \in \mathcal{N}_{\text{cov}}}{\operatorname{argmin}} w_b^\top (\Omega_0 + \delta \delta^\top) w_b.$$

Then the worst-case total-variation radius over the covariance-only Gaussian class satisfies

$$\rho_{\text{wc}}(\gamma; \mathcal{C}_\gamma^{\text{cov}, G}) = \sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - \mathcal{L}(T_{\text{cal}, \gamma}) \right\|_{\text{TV}} \longrightarrow 0 \quad \text{as } \gamma \rightarrow \infty,$$

where $v \sim N(\delta, \Omega_0 + H)$.

The proof is given in Appendix A.7. The condition $w_b^\top \Delta_{\text{obs}} w_b = 0$ is equivalent to $w_b^\top H_e w_b = 0$ for all observed environments with positive weight, because $H_e \succeq 0$. Thus, the observed covariance perturbations do not change the residual variance in direction w_b . Under this condition, Lemma 3.8 drives the worst-case projected test covariance perturbation to zero along the path b_γ . Since the mean is fixed, this aligns the folded-normal score distributions in total variation.

The two propositions should be read as sufficient structural conditions, not as a classification of shift types into cases where total-variation control succeeds or fails. They show that robust validation can become informative when the DRIG penalty controls a projected feature of the residual law that determines the score distribution. More general Gaussian shift classes would require additional assumptions ensuring simultaneous control of the relevant projected residual means and variances.

The numerical study in Chapter 4 uses the mean-only and covariance-only classes to illustrate settings in which robust validation is informative in a worst-case sense. It also compares this behaviour with the setting based on the full DRIG second-moment uncertainty set, where the Gaussian second-moment interval from Proposition 3.7 provides a different way to use the risk-level uncertainty set.

Chapter 4

Numerical Population Study

Chapter 3 derived the population objects needed to study the interaction between fixed-reference DRIG and robust validation. Along the DRIG path, each value of γ defines a predictor b_γ , and this predictor determines the residual laws, the absolute-residual score distributions, the total-variation radii used for robust validation, and the resulting coverages and interval widths. The theoretical analysis also showed that risk-level second-moment control and score-distribution control are distinct. The full second-moment uncertainty set may control projected residual moments, but it does not by itself yield informative worst-case total-variation control.

This chapter evaluates these mechanisms numerically at the population level. No finite samples are generated. All quantities are computed directly from the Gaussian residual and score laws derived in Chapter 3. The numerical study uses two complementary perspectives. In the *matched- γ analysis*, varying γ changes both the predictor b_γ and the admissible class of test shifts $\mathcal{A}_\gamma$. This is the guarantee-oriented perspective, because it evaluates the predictor and test class associated with the same value of γ . In the *fixed-target diagnostic*, the admissible class is held fixed, denoted by $\mathcal{A}_\eta$. Varying γ then changes only the predictor b_γ , which makes it possible to compare different points along the DRIG path against one common target class.

The first two settings consider the mean-only and covariance-only Gaussian subclasses from Section 3.6. These settings illustrate cases in which the DRIG path controls the part of the residual law that determines the conformal score distribution, so total-variation robust validation can become finite and informative. The third setting considers the full second-moment uncertainty set. There, the same DRIG path can still improve Gaussian moment-based intervals, but worst-case robust validation over the unrestricted full set remains vacuous.

4.1 Setup and evaluation criteria

All numerical settings use $p = 2$ covariates and nominal miscoverage level $\alpha = 0.1$. The DRIG path is evaluated on the grid $\gamma \in \{0, 0.01, \dots, 15\}$. Each setting contains a reference environment and two shifted observed environments. These shifted environments determine the observed heterogeneity used both to compute the DRIG path and to construct the admissible test classes. For each grid value, b_γ is computed from the fixed-reference objective in Section 3.2. The induced residual laws, score laws, total-variation radii, cover-

ages, and interval widths are then evaluated directly from the population formulas derived in Chapter 3. No finite samples are generated.

All three settings use the same structural matrix. The true coefficient is $\beta = (2, -0.75)^\top$, and the structural equations are $X = \xi_X$ and $Y = \beta^\top X + \xi_Y$. Thus, the structural noise vector has dimension $p + 1 = 3$. The observed environments are equally weighted. In the mean-only setting, the common covariance matrix is $\Omega = I_3$, and the noise means are

$$\delta^0 = 0, \quad \delta^1 = 0.75u, \quad \delta^2 = 1.50u, \quad u = (1, 1, 1)^\top.$$

Thus, the numbers 0, 0.75, and 1.50 are scalar multipliers of the common mean-shift direction u . In the covariance-only setting, the common noise mean is $\delta = (0.55, 0.15, 0.55)^\top$, and the covariance matrices are

$$\Omega^0 = I_3, \quad \Omega^1 = I_3 + 1.5uu^\top, \quad \Omega^2 = I_3 + 4.0uu^\top.$$

Here the numbers 0, 1.5, and 4.0 are scalar multipliers of the rank-one covariance perturbation $uu^\top$. The full second-moment setting combines these two types of heterogeneity. Its observed environments use the same mean-shift multipliers and covariance-perturbation multipliers in the direction u .

For each value of γ , the calibration score law is the weighted mixture

$$P_{\text{cal},\gamma}^T = \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e P_{e,\gamma}^T.$$

As before, $w_\gamma := w_{b_\gamma}$ denotes the residual direction induced by b_γ . Since the residual laws are Gaussian, the absolute-residual score laws are folded-normal distributions or mixtures of folded-normal distributions. The standard conformal interval uses the $1 - \alpha$ quantile of $P_{\text{cal},\gamma}^T$. It is therefore calibrated for the observed calibration mixture, but it is not corrected for a worst-case test shift.

The numerical study uses two complementary evaluation perspectives. The first is the matched- γ analysis, which is the guarantee-oriented perspective. In this analysis, the predictor and the admissible test class are indexed by the same value of γ . We write the admissible class generically as $\mathcal{A}_\gamma$. In the mean-only setting, $\mathcal{A}_\gamma = \mathcal{C}_\gamma^{\text{mean},G}$. In the covariance-only setting, $\mathcal{A}_\gamma = \mathcal{C}_\gamma^{\text{cov},G}$. In the full second-moment setting, $\mathcal{A}_\gamma = \mathcal{C}_\gamma$, where $\mathcal{C}_\gamma$ is the full second-moment uncertainty set determined by S_γ . If $P_{\gamma,P}^T$ denotes the absolute-residual score law induced by b_γ under a test distribution P , the corresponding worst-case total-variation radius is

$$\rho_{\text{wc}}(\gamma, \mathcal{A}_\gamma) = \sup_{P \in \mathcal{A}_\gamma} 2 \left\| P_{\gamma,P}^T - P_{\text{cal},\gamma}^T \right\|_{\text{TV}}.$$

The matched robust-validation quantile level is

$$u_\gamma^{\text{RV}} = \min \left\{ 1 - \alpha + \frac{\rho_{\text{wc}}(\gamma, \mathcal{A}_\gamma)}{2}, 1 \right\}.$$

The first grid value for which the matched robust-validation interval is finite is denoted by γ_{fin} . The grid value with the smallest finite matched robust-validation interval width is denoted by γ_{width}^* .

The second evaluation perspective used below is the fixed-target diagnostic. The fixed-target diagnostic keeps the admissible test class fixed at severity $\eta = 5$. This fixed class

is denoted by $\mathcal{A}_\eta$. In this diagnostic, varying γ changes only the predictor b_γ , while the target class stays fixed. Thus, the fixed-target diagnostic compares different points along the DRIG path against one common target class. The fixed-target robust-validation interval uses the same construction as above, but with $\rho_{\text{wc}}(\gamma, \mathcal{A}_\eta)$ instead of $\rho_{\text{wc}}(\gamma, \mathcal{A}_\gamma)$. The distinction between these two radii is important for interpreting the fixed-target figures below.

The fixed-target analysis also compares the value of γ selected by residual risk with the value selected by interval width. Let

$$R_{\text{wc},\eta}(\gamma) = \sup_{P \in \mathcal{A}_\eta} \mathbb{E}_P [U_\gamma^2]$$

denote the worst-case residual second moment over the fixed target class. The minimizer of this quantity on the grid is denoted by $\gamma_{\text{risk},\eta}^*$. The minimizer of the fixed-target interval width is denoted by $\gamma_{\text{width},\eta}^*$. These two values need not coincide, because interval width depends not only on the residual second moment, but also on the calibration score law and on the robust-validation correction.

Since $\alpha = 0.1$, the threshold $2\alpha = 0.2$ separates finite and vacuous total-variation robust-validation intervals. If the worst-case radius is at least 2α , then the robust-validation quantile level is clipped at 1. The folded-normal score laws have unbounded support, so the corresponding robust-validation interval has infinite width. In the figures, missing robust-validation curves correspond to such infinite intervals.

For the full second-moment setting, Gaussian moment intervals are also reported. These intervals are not robust-validation intervals. They do not use the calibration score law or a total-variation radius. Instead, they use the residual second-moment bound from Proposition 3.7. For a second-moment matrix S , the half-width is

$$q_{\text{mom}}(\gamma, S) = \sqrt{w_\gamma^\top S w_\gamma} \sqrt{1 + z_{1-\alpha/2}^2}.$$

The plotted interval width is twice this half-width. In the matched version, $S = S_\gamma$. In the fixed-target version, $S = S_\eta$. These intervals are included to show what second-moment information can provide under the Gaussian moment model, even though the same information is not sufficient for worst-case robust validation over the full set.

The total-variation radii are evaluated by one-dimensional numerical integration of folded-normal densities. In the mean-only subclass, the worst-case search reduces to a search over the projected residual mean. In the covariance-only subclass, it reduces to a search over the projected residual variance. For the full second-moment set, Proposition 3.6 implies that the worst-case total-variation radius is maximal, so total-variation robust validation is vacuous independently of the numerical grid.

4.2 Mean-only Gaussian subclass

The mean-only setting is the first structured case from Section 3.6.1. The observed environments differ through the mean of the SCM noise, while the covariance matrix is fixed. In this setting, changes in the residual law are driven by the projected residual mean $w_\gamma^\top \delta$. Proposition 3.9 shows that, when the mean-invariance set in (3.1) is nonempty, the DRIG penalty controls this projected mean shift. The numerical results in this section show how this predictor-side control translates into alignment of the absolute-residual score laws.

Figure 4.1 shows the fixed-target diagnostic. All curves are evaluated on the same target class $\mathcal{A}_\eta = \mathcal{C}_\eta^{\text{mean}, G}$. The standard conformal interval uses only the calibration score mixture. The matched robust-validation interval is calibrated for the class $\mathcal{C}_\gamma^{\text{mean}, G}$, which changes with γ . The fixed-target robust-validation interval is calibrated directly for $\mathcal{C}_\eta^{\text{mean}, G}$. Thus, the three curves answer different questions even though they are evaluated on the same fixed target class.

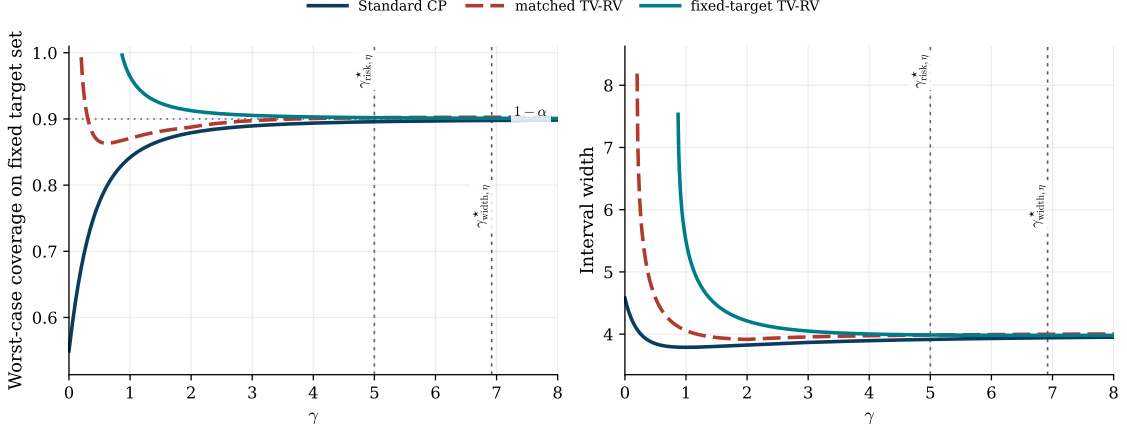


Figure 4.1: Fixed-target diagnostic in the mean-only class. The target set is fixed, while the predictor b_γ changes along the DRIG path. Left: worst-case coverage on the fixed target set. Right: interval widths. The matched robust-validation interval is calibrated for $\mathcal{C}_\gamma^{\text{mean}, G}$, whereas the fixed-target robust-validation interval is calibrated directly for the fixed target class. Missing robust-validation values correspond to infinite intervals.

The left panel shows that standard conformal prediction undercovers for small values of γ . As γ increases, the residual direction induced by the predictor becomes less sensitive to the admissible mean-shift directions, and the worst-case coverage improves. The fixed-target robust-validation interval gives a non-vacuous worst-case correction once its width is finite. The matched robust-validation interval can still undercover on the fixed target class when $\gamma < \eta$, because it is calibrated for $\mathcal{C}_\gamma^{\text{mean}, G}$, not for the fixed target class. Once $\gamma \geq \eta$, the matched class contains the fixed target class in this nested mean-only construction, so the matched robust-validation guarantee also applies to the fixed target class.

The right panel shows the corresponding width tradeoff. The fixed-target robust-validation interval is initially wide because it protects against the fixed target class even when the predictor is still close to the reference predictor. Its width decreases along the path and is minimized at $\gamma_{\text{width}, \eta}^* = 6.92$, which is larger than the fixed-target risk optimum $\gamma_{\text{risk}, \eta}^* = 5.00$. The comparison across settings is summarized in Table 4.1.

Figure 4.2 shows the matched- γ analysis. Here the predictor and the admissible class are indexed by the same value of γ . The left panel shows the worst-case total-variation radius over $\mathcal{C}_\gamma^{\text{mean}, G}$. At $\gamma = 0$, this radius is approximately 0.343. The matched robust-validation interval is therefore infinite at the reference predictor. The radius decreases rapidly along the path, first falls below the threshold at $\gamma_{\text{fin}} = 0.20$, and approaches zero for large γ in this mean-only example. The right panel shows that the smallest matched robust-validation interval width is attained at $\gamma_{\text{width}}^* = 2.00$, with width 3.92.

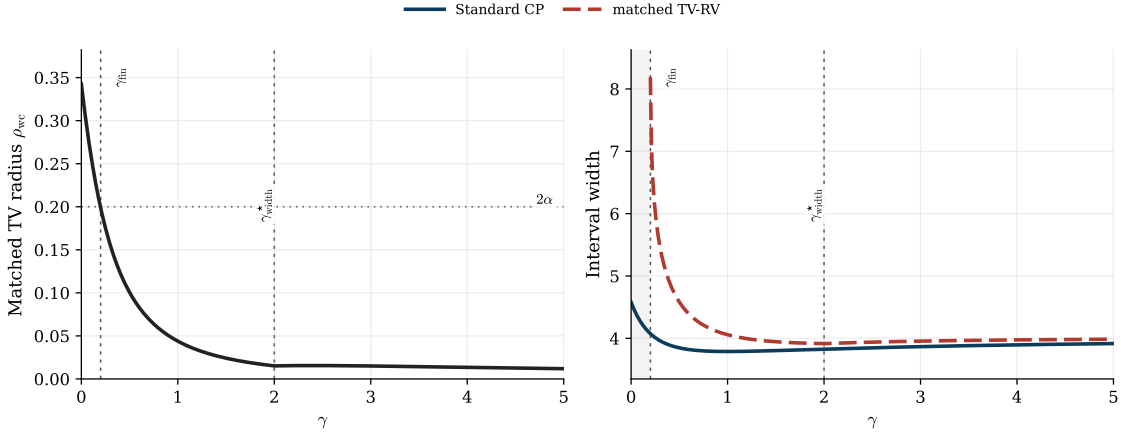


Figure 4.2: Matched- γ robust validation in the mean-only class. Left: worst-case radius over $\mathcal{C}_\gamma^{\text{mean},G}$, with the horizontal line marking $2\alpha = 0.2$. Right: standard conformal and matched robust-validation interval widths.

Although the two width panels look similar, they correspond to different criteria. Figure 4.1 evaluates all predictors against the fixed target class, whereas Figure 4.2 uses the matched class $\mathcal{C}_\gamma^{\text{mean},G}$. Hence $\gamma_{\text{width}}^* = 2.00$ is an optimum for the matched criterion, not for the fixed-target diagnostic.

Figure 4.3 illustrates the same mechanism at the level of score densities. The plotted densities are distributions of the absolute residual score. In each panel, the calibration score law is compared with the admissible test score law that maximizes the total-variation distance. At $\gamma = 0$, the two score laws are visibly separated. At $\gamma = \eta$, they are much more closely aligned. This alignment explains why the worst-case total-variation radius decreases along the DRIG path.

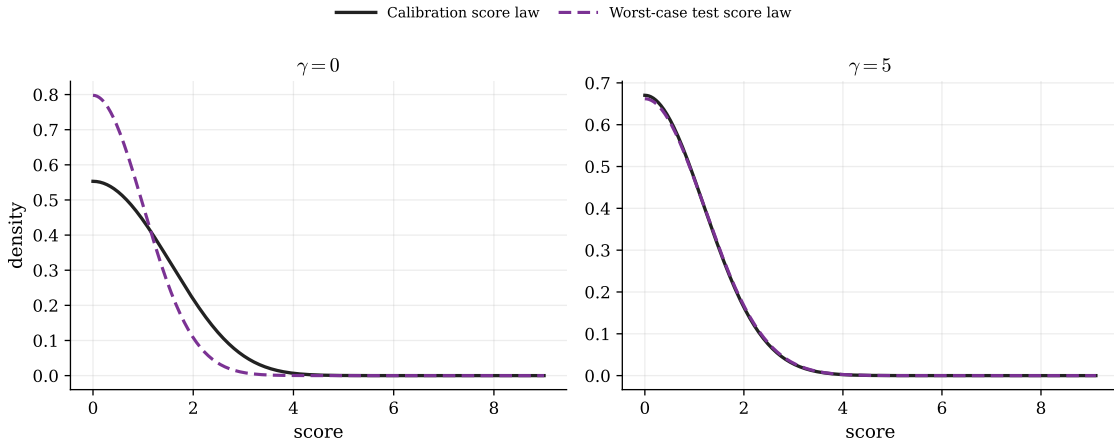


Figure 4.3: Score densities in the mean-only class. Each panel compares the calibration score law with the worst-case admissible test score law for the displayed value of γ . The score is the absolute residual. Alignment of these score laws is the distributional quantity needed for total-variation robust validation.

The mean-only setting is therefore a positive structured example of the connection between predictor-side and calibration-side robustness. The DRIG path reduces projected residual

mean shifts, and these projected shifts determine how the folded-normal score laws move in this subclass. The matched- γ analysis shows when robust validation becomes finite for the corresponding uncertainty set. The fixed-target diagnostic shows how the same predictors behave against one common target class. The covariance-only setting below has the same qualitative structure, but the score-relevant quantity changes from a projected mean to a projected variance.

4.3 Covariance-only Gaussian subclass

The covariance-only setting is the second structured case from Section 3.6.2. The noise mean is fixed, while the covariance matrix changes across environments. The qualitative question is the same as in the mean-only setting: does the DRIG path make total-variation robust validation finite and informative? The mechanism is different. In the mean-only setting, the score-relevant component is the projected residual mean. In the covariance-only setting, it is the projected residual variance $w_\gamma^\top \Omega_e w_\gamma$. Under the positive-semidefinite structure from Section 3.6.2, the DRIG penalty controls these projected variance perturbations.

Figure 4.4 applies the fixed-target diagnostic to the covariance-only target class $\mathcal{A}_\eta = \mathcal{C}_\eta^{\text{cov},G}$. The interpretation of the three interval types is the same as in Figure 4.1. Standard conformal prediction uses only the calibration score mixture. The matched robust-validation interval is calibrated for the class $\mathcal{C}_\gamma^{\text{cov},G}$, which changes with γ . The fixed-target robust-validation interval is calibrated directly for the fixed target class.

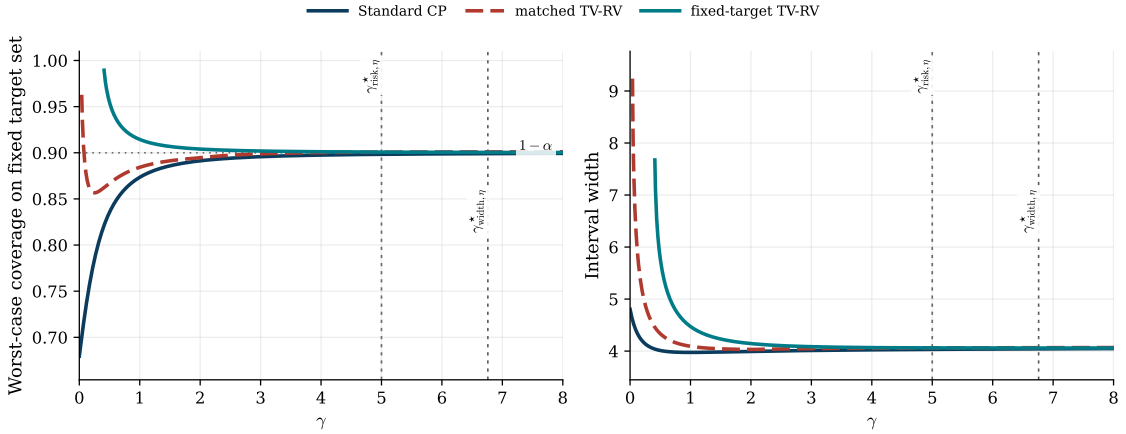


Figure 4.4: Fixed-target diagnostic in the covariance-only class. The target class is fixed, while b_γ varies along the DRIG path. Left: worst-case coverage under covariance shifts. Right: corresponding interval widths. The comparison mirrors Figure 4.1, but the projected score-relevant component is now the projected residual variance.

The qualitative behavior is analogous to the mean-only setting. Standard conformal prediction undercovers for small values of γ , and the fixed-target robust-validation interval gives a non-vacuous worst-case correction once its width is finite. The matched interval can again undercover on the fixed target class when $\gamma < \eta$, because its guarantee is for $\mathcal{C}_\gamma^{\text{cov},G}$, not for the fixed target class. Once $\gamma \geq \eta$, the matched covariance-only class contains the fixed target class in this nested construction, so the matched robust-validation guarantee also applies to the fixed target class.

The right panel shows the corresponding width tradeoff. The fixed-target robust-validation width decreases along the path and is minimized at $\gamma_{\text{width},\eta}^* = 6.76$. As in the mean-only setting, this value is slightly larger than the fixed-target risk optimum $\gamma_{\text{risk},\eta}^* = 5.00$. The reason is again that interval width depends on the calibration score law and on the robust-validation correction, not only on the fixed-target residual second moment. This similarity is important. The informative behavior of robust validation is not tied specifically to mean shifts. What matters is that the DRIG path controls the score-relevant projected component of the residual law.

Figure 4.5 shows the matched- γ analysis. The worst-case radius is now taken over $\mathcal{C}_\gamma^{\text{cov},G}$. At $\gamma = 0$, the radius is approximately 0.229, slightly above the threshold $2\alpha = 0.2$. Robust validation is therefore still vacuous at the reference predictor, but the radius is already close to the finite-interval threshold. On the numerical grid, the interval first becomes finite at $\gamma_{\text{fin}} = 0.04$. The smallest finite matched robust-validation interval width is again attained at $\gamma_{\text{width}}^* = 2.00$, with width approximately 4.03.

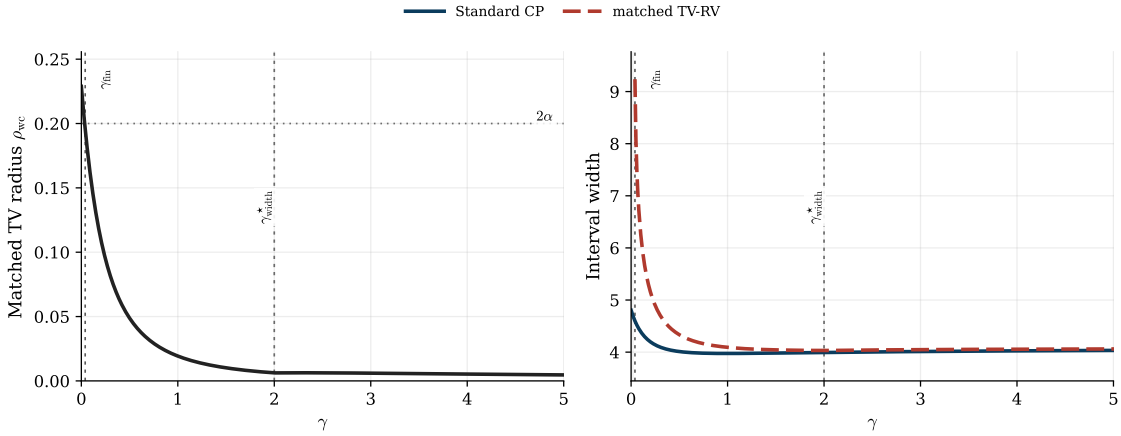


Figure 4.5: Matched- γ robust validation in the covariance-only class. Left: worst-case total-variation radius over $\mathcal{C}_\gamma^{\text{cov},G}$, with the horizontal line marking $2\alpha = 0.2$. Right: standard conformal and matched robust-validation interval widths. Missing robust-validation values correspond to infinite intervals.

The matched- γ diagnostic in Figure 4.5 is included to make the comparison with the mean-only case explicit. The qualitative shape is similar, but the initial radius is smaller and the first finite interval occurs earlier, at $\gamma_{\text{fin}} = 0.04$.

Figure 4.6 isolates the projected variance mechanism that distinguishes the covariance-only setting from the mean-only setting. The left panel shows the observed residual standard deviations

$$\sqrt{w_\gamma^\top \Omega_e w_\gamma}.$$

The covariance matrices themselves are not made equal along the path. Instead, the DRIG path changes the residual direction w_γ so that the observed residual variances become more similar after projection onto this direction. This projected variance determines the scale of the Gaussian residual and therefore the scale of the folded-normal score law.

The right panel uses the one-dimensional reduction of the covariance-only class from Section 3.6.2. In that class, an admissible test law has covariance matrix $\Omega_0 + H$, where the perturbation satisfies $0 \preceq H \preceq \gamma \Delta_{\text{obs}}$. After projection onto the residual direction, this

covariance perturbation contributes the additional residual variance

$$h = w_\gamma^\top H w_\gamma.$$

The positive-semidefinite order implies

$$0 \leq h \leq h_{\max}(\gamma), \quad h_{\max}(\gamma) = \gamma w_\gamma^\top \Delta_{\text{obs}} w_\gamma.$$

The curve $h_{\max}(\gamma)$ can first increase because the admissible class grows with γ . It can later decrease because the factor $w_\gamma^\top \Delta_{\text{obs}} w_\gamma$ becomes small. The total-variation maximizing value of h need not always be the largest admissible perturbation. In this numerical example, the maximizer switches from the lower endpoint to the upper endpoint along the path.

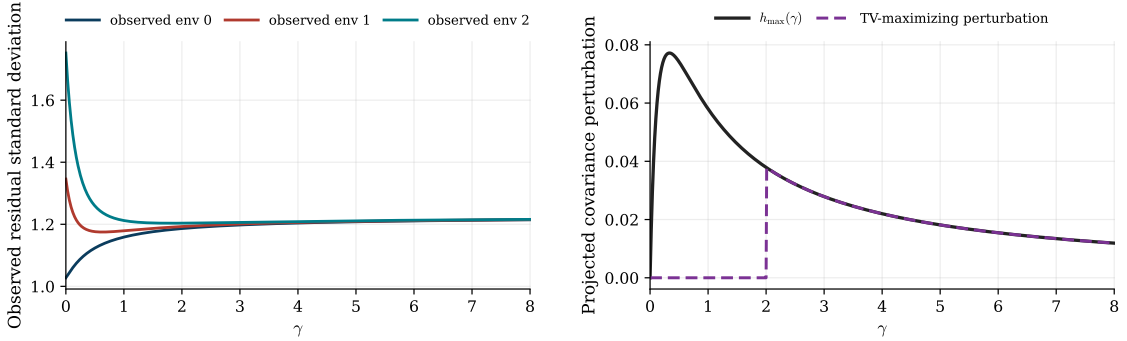


Figure 4.6: Mechanism plot for the covariance-only class. Left: observed residual standard deviations along the DRIG path. Right: largest admissible projected covariance perturbation and total-variation maximizing projected perturbation. The covariance matrices are not made equal, but their projections onto the residual direction become more aligned.

The covariance-only setting therefore complements the mean-only setting. Both settings show that total-variation robust validation can become finite and informative along the DRIG path. The common reason is score-level structure. In the mean-only class, the relevant score-level change is induced by projected mean shifts. In the covariance-only class, it is induced by projected variance shifts. This prepares the comparison with the full second-moment uncertainty set, where no such single projected component is controlled by the set alone.

4.4 Full second-moment uncertainty set

The third setting evaluates the full second-moment uncertainty set. In contrast to the mean-only and covariance-only subclasses, this set does not isolate a single score-relevant projected component of the residual law. It constrains the second moment of the shifted noise vector, but it does not separately control how the conformal score law changes. This setting therefore illustrates the limitation identified in Proposition 3.6: moment control and total-variation control of the score distribution are different requirements.

Figure 4.7 shows the robust-validation quantities for the full set. The worst-case total-variation radius is equal to 2 for all values of γ . Since $2\alpha = 0.2$, the robust-validation quantile level is clipped at 1 throughout the path. The corresponding robust-validation interval has infinite width because the score laws have unbounded support. Thus, robust

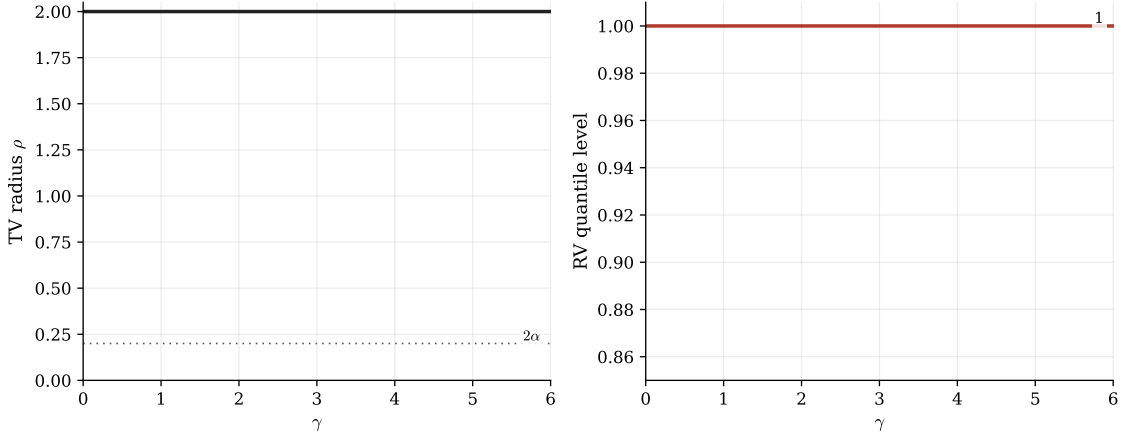


Figure 4.7: Vacuity of robust validation over the full second-moment uncertainty set. Left: the full-set worst-case total-variation radius is maximal for all values of γ . Right: the corresponding robust-validation quantile level is clipped at 1, so the robust-validation interval has infinite width.

validation over the full second-moment set is vacuous, both in the matched- γ analysis and in the fixed-target diagnostic.

This result should not be read as a statement that simultaneous mean and covariance shifts are inherently impossible to handle. The point is more specific. The unrestricted full second-moment set is too broad for worst-case robust validation, because it controls projected second moments but does not control the score law itself. The structured subclasses in Sections 4.2 and 4.3 become informative only because additional score-level structure is available.

Although robust validation is vacuous in the full set, the same DRIG path can still improve Gaussian moment intervals. These intervals use the bound from Proposition 3.7. For a given second-moment matrix S , their half-width is proportional to

$$\sqrt{w_\gamma^\top S w_\gamma}.$$

They therefore depend directly on the projected residual second moment. In the fixed-target version, the evaluated matrix is S_η . In the matched- γ version, it is S_γ . These intervals do not use the calibration score law and are not robust-validation intervals.

Figure 4.8 shows the corresponding moment-based quantities. The left panel reports interval widths based on S_η and S_γ , together with the standard conformal width as a calibration-based reference. The right panel shows the projected second moments that determine the Gaussian moment widths, as well as the maximum observed risk along the path. The fixed-target Gaussian moment width decreases strongly along the path and is minimized at $\gamma = 5.00$. The projected second moment in the right panel has the same minimizer. This coincidence occurs because, for these intervals, the width is a monotone function of the projected second moment.

The matched moment curve has a different interpretation, because both the predictor and the second-moment matrix change with γ . For larger values of γ , the matched and fixed-target moment widths can nevertheless be close. The reason is that the additional

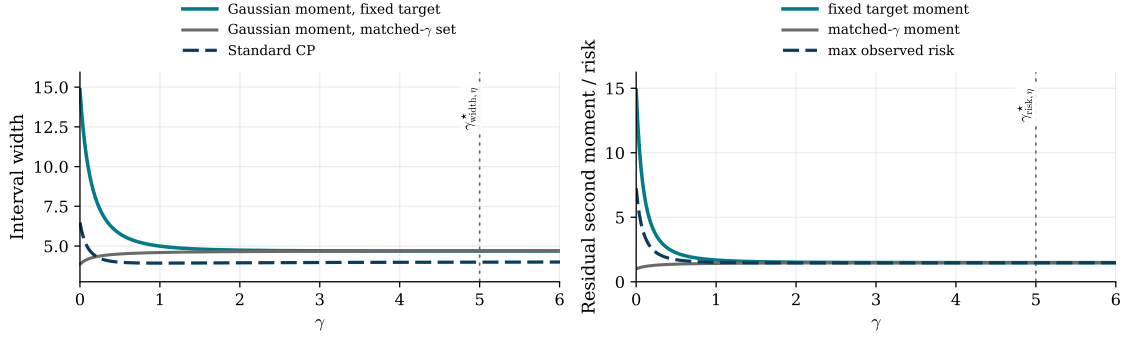


Figure 4.8: Gaussian moment intervals for the full second-moment uncertainty set. Left: interval widths based on the fixed target matrix S_η and on the matched matrix S_γ , together with the standard conformal width. Right: the corresponding projected second moments and the maximum observed risk along the path b_γ . These are moment-based Gaussian intervals, not robust-validation intervals.

second-moment contribution is projected through w_γ . If $w_\gamma^\top \Delta_{\text{obs}} w_\gamma$ is small, increasing the size of the second-moment set has little effect in the residual direction.

The Gaussian moment intervals also clarify why the full second-moment setting should not be summarized simply as a failure case. Moment information can be useful when the goal is a Gaussian moment-based prediction interval. Such intervals may also be competitive in the structured Gaussian subclasses. This does not contradict the robust-validation results. The two approaches use different information and give different guarantees. The moment intervals rely on Gaussian second-moment structure. Robust validation instead works through the calibration score law and a total-variation correction, so it targets robustness of the conformal score distribution directly.

The full second-moment setting therefore separates two notions of robustness. The DRIG path can reduce projected residual second moments and thereby improve moment-based intervals. However, without additional assumptions controlling the score distribution, this does not yield informative robust validation. The positive results in the mean-only and covariance-only subclasses should therefore be interpreted as sufficient score-level conditions, not as evidence that only pure mean shifts or pure covariance shifts can be handled.

4.5 Summary

The numerical population study supports the distinction developed in Chapter 3. The mean-only and covariance-only subclasses show similar qualitative behavior. In both cases, the DRIG path controls a projected feature of the residual law that determines the conformal score distribution. In the mean-only class, this feature is the projected residual mean. In the covariance-only class, it is the projected residual variance. Because of this additional score-level structure, the matched total-variation radius falls below the threshold 2α , and the matched robust-validation interval becomes finite.

The full second-moment setting shows why this conclusion does not follow from moment control alone. The DRIG path can reduce projected residual second moments and thereby shorten Gaussian moment intervals. However, the unrestricted full second-moment set does not control the score law itself. The worst-case total-variation radius remains maxi-

mal, so robust validation over the full set stays vacuous throughout the path.

Table 4.1 summarizes the main numerical quantities. The columns $\rho_{wc}(0)$, γ_{fin} , and γ_{width}^* refer to the matched- γ robust-validation analysis. The columns $\gamma_{risk,\eta}^*$ and $\gamma_{width,\eta}^*$ refer to the fixed-target diagnostic. For the full second-moment setting, the final width optimum refers to the Gaussian moment interval, not to a total-variation robust-validation interval.

Table 4.1: Summary of the numerical population results on the γ -grid. The matched columns report the initial worst-case total-variation radius, the first finite robust-validation interval, and the smallest finite matched robust-validation width. The fixed-target columns report the grid values minimizing fixed-target residual risk and fixed-target interval width. In the full second-moment setting, robust validation is vacuous, and the final width optimum refers to the Gaussian moment interval.

Setting	Score-level control	$\rho_{wc}(0)$	γ_{fin}	γ_{width}^*	$\gamma_{risk,\eta}^*$	$\gamma_{width,\eta}^*$
Mean-only	projected mean	0.343	0.20	2.00	5.00	6.92
Covariance-only	projected variance	0.229	0.04	2.00	5.00	6.76
Full second-moment	not controlled by set alone	2.00	—	—	5.00	5.00

The fixed-target risk optimum occurs at $\gamma = 5.00$ in all three numerical settings. This is natural in the present design, because the fixed target class is constructed at the same severity. The value $\gamma = 5.00$ therefore aligns the DRIG path with the severity used to define the fixed target. The width optimum need not coincide with this value. In the two structured subclasses, the fixed-target robust-validation width is minimized at a slightly larger value of γ , because the width also depends on the calibration score law and on the total-variation correction. In the full second-moment setting, the fixed-target Gaussian moment width is minimized at the same value as the fixed-target residual risk, because this width is a monotone function of the projected second moment.

The numerical study therefore shows when predictor-side robustness becomes useful for calibration-side robustness. DRIG controls residual risk along a path of predictors. Robust validation requires control of the conformal score law. The two are connected in the structured subclasses because the DRIG penalty controls the projected residual feature that determines the score distribution. They are not connected for the unrestricted full second-moment set without additional score-level assumptions.

The value of γ should therefore be interpreted relative to the criterion of interest. Matched robust-validation width, fixed-target residual risk, fixed-target robust-validation width, and Gaussian moment width can select different points along the same DRIG path. This criterion dependence is not a defect of the numerical example. The numerical results therefore support the following conclusion: risk-level robustness, score-distribution robustness, and interval efficiency are connected only when the admissible shifts impose enough structure on the score law.

Chapter 5

Discussion

This thesis studies conformal prediction under distribution shift, with a focus on the interaction between predictor-side robustness and calibration-side robustness. The central observation is that these two forms of robustness meet at the level of conformal scores. Robust validation adjusts the calibration threshold for a fixed predictor. DRIG changes the predictor and therefore changes the residual and score distributions before calibration is applied. Hence, a predictor that is more stable under environment shifts can also change the amount of calibration-side protection needed for conformal prediction.

The main contribution is the worked population analysis developed in Chapter 3 and evaluated numerically in Chapter 4. The analysis is carried out in a fixed- B Gaussian structural equation model. The fixed-reference DRIG objective yields an explicit coefficient path b_γ . Along this path, the residual laws are Gaussian and the absolute residual scores have folded-normal distributions. This makes it possible to compute calibration score laws, admissible test score laws, total-variation robust-validation radii, coverages, and interval widths as population quantities. The analysis isolates the mechanism

$$\gamma \longmapsto b_\gamma \longmapsto F_{\text{cal},\gamma}, P_{v,\gamma}^T \longmapsto \rho_{\text{wc}}(\gamma) \longmapsto \text{Width}_\gamma^{\text{RV}}.$$

The analysis is designed as a mechanism study. It makes explicit how changing the predictor affects the score distributions to which robust validation is applied.

Chapter 4 uses two evaluation perspectives to separate guarantee statements from predictor comparisons. The matched- γ perspective evaluates the pair consisting of b_γ and the admissible class indexed by the same value of γ . It is therefore the guarantee-oriented perspective. The fixed-target diagnostic instead keeps one target class fixed and varies only the predictor. This separates changes in the predictor from changes in the uncertainty class. In the numerical examples, the fixed-target residual risk and the fixed-target robust-validation width are minimized at different values of γ . This shows that the DRIG parameter should not be interpreted as having one universally optimal value. Matched robust-validation width, fixed-target risk, fixed-target robust-validation width, and Gaussian moment width may select different points along the same coefficient path.

Under these two perspectives, the numerical population study shows that the DRIG path can make robust validation finite and informative when the shift structure controls the score-relevant part of the residual law. In the mean-only Gaussian subclass, this happens through projected mean alignment. In the covariance-only Gaussian subclass, it happens through projected variance alignment. In both cases, the matched total-variation radius

falls below the critical threshold 2α , so the matched robust-validation interval becomes finite and provides a non-vacuous worst-case coverage guarantee over the corresponding structured uncertainty set. In one case, the relevant score-level change is caused by a residual mean shift. In the other case, it is caused by a residual variance shift.

This distinction is important for interpreting the positive results. The numerical study should not be read as saying that mean shifts and covariance shifts are the only cases in which robust validation can become informative. Rather, these subclasses provide two tractable examples in which the risk-level DRIG penalty can be translated into control of a score-relevant one-dimensional quantity. What matters is not the label of the shift type itself, but whether the admissible shifts impose enough structure on the induced residual and score laws. In the mean-only subclass, controlling the projected mean is enough because the residual variance is fixed. In the covariance-only subclass, controlling the projected variance is enough because the residual mean is fixed. More general shift classes would require a joint condition controlling both components, or another form of score-level control.

This dependence on the chosen criterion is not just a numerical accident. The different criteria measure different objects. Prediction risk depends on the second moment of the residual. Standard conformal width depends on the calibration score distribution induced by the predictor. Robust-validation width depends additionally on the discrepancy between calibration and admissible test score laws. Thus, a predictor that is good for residual risk need not be optimal for robust conformal interval width. Conversely, a value of γ that gives short robust-validation intervals for a matched uncertainty class need not be optimal for a fixed target class. The numerical results therefore support the view that choosing γ should be tied to the intended robustness criterion.

The analysis also clarifies a limitation of using the full second-moment uncertainty set directly for robust validation. This uncertainty set gives residual moment control and is natural for risk-level statements, but it does not control the total-variation distance between score laws. It can contain future laws with maximal total-variation discrepancy from the calibration score law, making worst-case robust validation over the unrestricted full set vacuous. This does not mean that simultaneous mean and covariance shifts are inherently impossible to handle. It means that the full second-moment constraint is too broad for total-variation robust validation unless additional score-level structure is imposed.

The Gaussian moment intervals in the full second-moment setting illustrate a different consequence of the same second-moment information. They can become substantially shorter along the fixed-reference path because their width depends directly on the projected residual second moment. However, these intervals are not robust-validation intervals. They do not use the calibration score law and they do not control a discrepancy between calibration and test score distributions. They answer a moment-based Gaussian prediction question, while robust validation answers a score-distribution calibration question. Thus, the fact that Gaussian moment intervals can be useful does not contradict the vacuity of total-variation robust validation over the full set.

In a Gaussian population model, moment-based intervals can be attractive because the residual second moment is analytically accessible and directly tied to the interval width. Robust validation has a different role. It is formulated at the level of score distributions, which is the level at which conformal calibration operates. This makes it conceptually closer to conformal prediction, and potentially more transferable to settings where

Gaussian moment assumptions are not appropriate. The price is that informative robust validation requires control of the score law, not only of residual moments.

Several limitations remain. First, the worked example uses a fixed-reference specialization of the DRIG objective. This choice makes the population path analytically tractable and preserves the risk-contrast mechanism relevant for the score-level analysis, but it is not the full minimum-risk DRIG objective. A direct analysis of the original objective would require handling its piecewise quadratic structure and the possible change of the minimum-risk environment along the path. Such an analysis could show whether the mechanisms identified here persist for the original DRIG path, or whether the fixed-reference specialization changes the score-level behavior in important ways.

Second, the analysis is entirely population-level. The calibration score laws, admissible test score laws, robust-validation radii, coverages, and widths are computed from the known data-generating mechanism. This removes estimation error and makes the mechanism transparent, but it does not provide a finite-sample procedure for selecting γ or estimating the robust-validation radius in practice. In applications, both the predictor and the radius would need to be estimated from data, and the additional uncertainty could affect coverage and interval width. A practical procedure would also have to decide which class of shifts should be protected and how conservative the resulting intervals should be.

Third, the numerical study considers structured Gaussian shifts. These settings are useful because they make the residual and score laws explicit and interpretable. They are not intended to cover all possible distribution shifts, and the results do not imply that increasing the DRIG parameter always improves robust validation. What matters is whether the DRIG penalty controls a projected feature of the residual law that is relevant for the conformal score distribution. More general Gaussian shift classes would require assumptions that control projected residual means and variances jointly. Beyond Gaussian models, one would need conditions that control the score law directly.

Fourth, the robust-validation part of the worked example focuses on total variation. This choice gives a simple radius correction and a clear finite-versus-vacuous threshold at 2α , but it can be conservative. Other f -divergences could lead to different trade-offs between protection and interval width. They may also interact differently with changes in the score law along the DRIG path. Studying whether alternative divergences yield less conservative but still interpretable robust-validation intervals would be a natural extension.

Future work could proceed in several directions. One direction is to analyze the original DRIG objective rather than the fixed-reference specialization. Another is to develop data-driven procedures for choosing γ with the goal of minimizing robust-validation width subject to coverage. A third direction is to combine the population analysis with finite-sample estimation of the robust-validation radius. Further work could also study joint mean-covariance subclasses with explicit score-level conditions, compare different divergence choices, and test the mechanisms on real heterogeneous datasets.

Overall, the thesis shows that robust prediction and robust calibration should not be viewed as independent layers. Changing the predictor changes the score distributions that determine the robust-validation radius and interval width. DRIG can therefore make robust validation less conservative in structured shift settings, but this effect is mechanism-dependent. It occurs when the predictor-side penalty controls the part of the residual law that determines the conformal score distribution. It cannot be inferred from risk-level robustness alone.

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Appendix A

Proofs

A.1 Proof of Proposition 3.1

Fix $e \in \mathcal{E}_{\text{obs}}$ and $b \in \mathbb{R}^p$. By definition,

$$\ell_b = \begin{pmatrix} -b \\ 1 \end{pmatrix} \in \mathbb{R}^{p+1}, \quad Z^e = \begin{pmatrix} X^e \\ Y^e \end{pmatrix}.$$

Therefore the residual of the linear predictor $x \mapsto b^\top x$ is

$$Y^e - b^\top X^e = \ell_b^\top Z^e.$$

Hence

$$\begin{aligned} R_e(b) &= \mathbb{E} \left[(Y^e - b^\top X^e)^2 \right] \\ &= \mathbb{E} \left[(\ell_b^\top Z^e)^2 \right] \\ &= \mathbb{E} \left[\ell_b^\top Z^e (Z^e)^\top \ell_b \right]. \end{aligned}$$

Since ℓ_b is deterministic and $\Lambda_e = \mathbb{E}[Z^e (Z^e)^\top]$, this gives

$$R_e(b) = \ell_b^\top \Lambda_e \ell_b.$$

It remains to expand this expression using the block decomposition

$$\Lambda_e = \begin{pmatrix} \Lambda_{XX}^e & \Lambda_{XY}^e \\ (\Lambda_{XY}^e)^\top & \Lambda_{YY}^e \end{pmatrix}.$$

We have

$$\begin{aligned} \ell_b^\top \Lambda_e \ell_b &= \begin{pmatrix} -b^\top & 1 \end{pmatrix} \begin{pmatrix} \Lambda_{XX}^e & \Lambda_{XY}^e \\ (\Lambda_{XY}^e)^\top & \Lambda_{YY}^e \end{pmatrix} \begin{pmatrix} -b \\ 1 \end{pmatrix} \\ &= b^\top \Lambda_{XX}^e b - b^\top \Lambda_{XY}^e - (\Lambda_{XY}^e)^\top b + \Lambda_{YY}^e. \end{aligned}$$

The two middle terms are equal because they are real scalars. Thus

$$\ell_b^\top \Lambda_e \ell_b = b^\top \Lambda_{XX}^e b - 2b^\top \Lambda_{XY}^e + \Lambda_{YY}^e.$$

Combining this with $R_e(b) = \ell_b^\top \Lambda_e \ell_b$ yields

$$R_e(b) = \ell_b^\top \Lambda_e \ell_b = b^\top \Lambda_{XX}^e b - 2b^\top \Lambda_{XY}^e + \Lambda_{YY}^e.$$

This completes the proof.

A.2 Proof of Proposition 3.2

By definition,

$$L_\gamma^{\text{ref}}(b) = R_0(b) + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (R_e(b) - R_0(b)).$$

Since $\sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e = 1$, this can be written as

$$L_\gamma^{\text{ref}}(b) = (1 - \gamma)R_0(b) + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e R_e(b).$$

Using Proposition 3.1, for each $e \in \mathcal{E}_{\text{obs}}$,

$$R_e(b) = b^\top \Lambda_{XX}^e b - 2b^\top \Lambda_{XY}^e + \Lambda_{YY}^e.$$

Therefore,

$$\begin{aligned} L_\gamma^{\text{ref}}(b) &= (1 - \gamma) \left(b^\top \Lambda_{XX}^0 b - 2b^\top \Lambda_{XY}^0 + \Lambda_{YY}^0 \right) \\ &\quad + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e \left(b^\top \Lambda_{XX}^e b - 2b^\top \Lambda_{XY}^e + \Lambda_{YY}^e \right). \end{aligned}$$

Collecting the quadratic, linear, and constant terms gives

$$L_\gamma^{\text{ref}}(b) = b^\top A_\gamma b - 2b^\top h_\gamma + c_\gamma,$$

where

$$\begin{aligned} A_\gamma &= \Lambda_{XX}^0 + \gamma \left(\bar{\Lambda}_{XX, \text{obs}} - \Lambda_{XX}^0 \right), \\ h_\gamma &= \Lambda_{XY}^0 + \gamma \left(\bar{\Lambda}_{XY, \text{obs}} - \Lambda_{XY}^0 \right), \\ c_\gamma &= \Lambda_{YY}^0 + \gamma \left(\bar{\Lambda}_{YY, \text{obs}} - \Lambda_{YY}^0 \right). \end{aligned}$$

This proves the stated quadratic representation.

Assume now that $A_\gamma \succ 0$. Then L_γ^{ref} is a strictly convex and coercive quadratic function of b , because its Hessian is $2A_\gamma$. Hence it has a unique minimizer. Its gradient is

$$\nabla_b L_\gamma^{\text{ref}}(b) = 2A_\gamma b - 2h_\gamma.$$

The first-order condition is therefore $A_\gamma b = h_\gamma$. Since $A_\gamma \succ 0$, the matrix A_γ is invertible, and the unique solution is

$$b_\gamma = A_\gamma^{-1} h_\gamma.$$

Thus b_γ is the unique minimizer of L_γ^{ref} .

It remains to identify the cases $\gamma = 0$ and $\gamma = 1$. For $\gamma = 0$,

$$A_0 = \Lambda_{XX}^0, \quad h_0 = \Lambda_{XY}^0.$$

Hence

$$b_0 = A_0^{-1} h_0 = (\Lambda_{XX}^0)^{-1} \Lambda_{XY}^0 = b_{\text{ref}}^{\text{OLS}}.$$

For $\gamma = 1$,

$$A_1 = \bar{\Lambda}_{XX, \text{obs}}, \quad h_1 = \bar{\Lambda}_{XY, \text{obs}}.$$

Therefore

$$b_1 = A_1^{-1} h_1 = \bar{\Lambda}_{XX, \text{obs}}^{-1} \bar{\Lambda}_{XY, \text{obs}} = b_{\text{pld}}^{\text{OLS}}.$$

This completes the proof.

A.3 Proof of Proposition 3.6

We construct a future noise distribution satisfying the second-moment constraint whose induced score law is a point mass. Let

$$C := w^\top S w > 0$$

and define

$$q := \frac{S w}{\sqrt{C}}.$$

Then

$$w^\top q = \frac{w^\top S w}{\sqrt{C}} = \sqrt{C}.$$

We claim that $q q^\top \preceq S$. To see this, let $x \in \mathbb{R}^{p+1}$. Since $S \succeq 0$, the map $(u, v)_S := u^\top S v$ defines a semi-inner product. By the Cauchy–Schwarz inequality for this semi-inner product,

$$(x^\top S w)^2 \leq (x^\top S x)(w^\top S w) = C x^\top S x.$$

Therefore,

$$x^\top q q^\top x = (x^\top q)^2 = \frac{(x^\top S w)^2}{C} \leq x^\top S x.$$

Since this holds for every $x \in \mathbb{R}^{p+1}$, it follows that $q q^\top \preceq S$.

Now let P_v be the degenerate distribution that assigns probability one to q . Then

$$\mathbb{E}_{P_v}[v v^\top] = q q^\top \preceq S,$$

so P_v is feasible for the supremum in Proposition 3.6. Under this distribution,

$$|w^\top v| = |w^\top q| = \sqrt{C}$$

almost surely. Hence the induced score law is

$$\mathcal{L}(|w^\top v|) = \delta_{\sqrt{C}},$$

the point mass at $\sqrt{C}$.

By assumption, the calibration score distribution P_{cal}^T is continuous on $[0, \infty)$. In particular, $P_{\text{cal}}^T(\{\sqrt{C}\}) = 0$. Using the definition of total variation distance as a supremum over measurable sets,

$$\begin{aligned} \left\| \delta_{\sqrt{C}} - P_{\text{cal}}^T \right\|_{\text{TV}} &\geq \left| \delta_{\sqrt{C}}(\{\sqrt{C}\}) - P_{\text{cal}}^T(\{\sqrt{C}\}) \right| \\ &= 1. \end{aligned}$$

Total variation distance is always at most one. Therefore,

$$\left\| \delta_{\sqrt{C}} - P_{\text{cal}}^T \right\|_{\text{TV}} = 1.$$

Consequently,

$$2 \left\| \mathcal{L}(|w^\top v|) - P_{\text{cal}}^T \right\|_{\text{TV}} = 2.$$

Since $2\|\cdot\|_{\text{TV}}$ is always bounded above by 2, this feasible choice attains the largest possible value. Hence

$$\sup_{P_v: \mathbb{E}[v v^\top] \preceq S} 2 \left\| \mathcal{L}(|w^\top v|) - P_{\text{cal}}^T \right\|_{\text{TV}} = 2.$$

This completes the proof.

A.4 Proof of Proposition 3.7

Let $Z \sim N(0, 1)$. It suffices to prove the claim for the representation

$$Y = m + \sigma Z.$$

By the Cauchy–Schwarz inequality applied to (m, σ) and $(1, Z)$,

$$|m + \sigma Z| \leq \sqrt{m^2 + \sigma^2} \sqrt{1 + Z^2}.$$

Using $m^2 + \sigma^2 \leq M$, we obtain

$$|Y| \leq \sqrt{M} \sqrt{1 + Z^2}$$

almost surely under this representation. Therefore, on the event $|Z| \leq z_{1-\alpha/2}$, it holds that

$$|Y| \leq \sqrt{M} \sqrt{1 + z_{1-\alpha/2}^2}.$$

Since

$$\mathbb{P}(|Z| \leq z_{1-\alpha/2}) = 1 - \alpha,$$

the desired inequality follows. This completes the proof.

A.5 Proof of Lemma 3.8

Since b_γ minimizes $a + \gamma g$, and since $g(b_\infty) = 0$, we have

$$a(b_\gamma) + \gamma g(b_\gamma) \leq a(b_\infty) + \gamma g(b_\infty) = a(b_\infty).$$

As $g(b_\gamma) \geq 0$, this implies

$$a(b_\gamma) \leq a(b_\infty).$$

Since $a(b) \rightarrow \infty$ as $\|b\| \rightarrow \infty$, the family $(b_\gamma)_{\gamma \geq 1}$ is bounded.

Since a is continuous and coercive, it is bounded from below. Let

$$a_{\min} := \inf_{b \in \mathbb{R}^p} a(b) > -\infty.$$

The same inequality then gives

$$0 \leq g(b_\gamma) \leq \frac{a(b_\infty) - a_{\min}}{\gamma} \rightarrow 0.$$

Let b^* be an accumulation point of the family $(b_\gamma)_{\gamma \geq 1}$. Since $g(b_\gamma) \rightarrow 0$ and g is continuous, we have $g(b^*) = 0$. Thus $b^* \in \mathcal{N}$.

We now show that b^* minimizes a over $\mathcal{N}$. Let $b \in \mathcal{N}$. Since b_γ minimizes $a + \gamma g$, and since $g(b) = 0$, we have

$$a(b_\gamma) + \gamma g(b_\gamma) \leq a(b) + \gamma g(b) = a(b).$$

Again using $g(b_\gamma) \geq 0$, it follows that

$$a(b_\gamma) \leq a(b).$$

Passing to a convergent subsequence with limit b^* , and using continuity of a , yields

$$a(b^*) \leq a(b).$$

Since this holds for every $b \in \mathcal{N}$, the accumulation point b^* minimizes a over $\mathcal{N}$.

A nonnegative quadratic function has positive semidefinite Hessian and is therefore convex. Hence $\mathcal{N} = \{b : g(b) = 0\}$, the set of global minimizers of g , is convex. Since a is strictly convex, its minimizer over $\mathcal{N}$ is unique. By definition, this minimizer is b_∞ . Therefore every accumulation point of $(b_\gamma)_{\gamma \geq 1}$ equals b_∞ . Since the family is bounded, this implies

$$b_\gamma \rightarrow b_\infty.$$

It remains to prove the rate

$$\gamma g(b_\gamma) \rightarrow 0.$$

Since g is a nonnegative quadratic function and $g(b_\infty) = 0$, the point b_∞ is a global minimizer of g . Hence

$$\nabla g(b_\infty) = 0.$$

Because g is quadratic, its Taylor expansion around b_∞ is exact, and

$$g(b_\gamma) = \frac{1}{2}(b_\gamma - b_\infty)^\top \nabla^2 g(b_\infty)(b_\gamma - b_\infty).$$

The first-order optimality condition for b_γ is

$$\nabla a(b_\gamma) + \gamma \nabla g(b_\gamma) = 0.$$

Combining the previous two displays gives

$$\gamma g(b_\gamma) = -\frac{1}{2}(b_\gamma - b_\infty)^\top \nabla a(b_\gamma)(b_\gamma - b_\infty).$$

Since ∇a is continuous and $b_\gamma \rightarrow b_\infty$, the family $(\nabla a(b_\gamma))_{\gamma \geq 1}$ is bounded. Hence

$$\gamma g(b_\gamma) \rightarrow 0.$$

This completes the proof.

A.6 Proof of Proposition 3.9

In the mean-only setting, the residual of a linear predictor b in environment e satisfies

$$U^e(b) = w_b^\top \xi^e \sim N(w_b^\top \delta^e, w_b^\top \Omega w_b).$$

Since $\delta^0 = 0$ in the baseline environment, the fixed-reference objective becomes

$$\begin{aligned} L_\gamma^{\text{ref}}(b) &= R_0(b) + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (R_e(b) - R_0(b)) \\ &= w_b^\top \Omega w_b + \gamma w_b^\top M_{\text{obs}} w_b \\ &= a(b) + \gamma g(b), \end{aligned}$$

where

$$a(b) := w_b^\top \Omega w_b, \quad g(b) := w_b^\top M_{\text{obs}} w_b.$$

The function g is a nonnegative quadratic function, because $M_{\text{obs}} \succeq 0$. Moreover, a is a strictly convex quadratic function of b . Indeed, $K\Omega K^\top \succ 0$, and the Hessian of a with respect to b is twice the upper-left $p \times p$ block of $K\Omega K^\top$, hence is positive definite. In particular, $a(b) \rightarrow \infty$ whenever $\|b\| \rightarrow \infty$.

By the assumption of the proposition,

$$\mathcal{N}_{\text{mean}} = \{b \in \mathbb{R}^p : g(b) = 0\}$$

is nonempty, and

$$b_\infty^{\text{mean}} = \underset{b \in \mathcal{N}_{\text{mean}}}{\operatorname{argmin}} a(b).$$

Thus Lemma 3.8 applies. Hence

$$b_\gamma \rightarrow b_\infty^{\text{mean}}$$

and

$$\gamma g(b_\gamma) = \gamma w_\gamma^\top M_{\text{obs}} w_\gamma \rightarrow 0.$$

We now use this rate to control the observed residual means and the admissible test residual means. For any observed environment e with $\omega^e > 0$,

$$g(b_\gamma) = w_\gamma^\top M_{\text{obs}} w_\gamma = \sum_{e' \in \mathcal{E}_{\text{obs}}} \omega^{e'} (w_\gamma^\top \delta^{e'})^2 \geq \omega^e (w_\gamma^\top \delta^e)^2.$$

Hence

$$(w_\gamma^\top \delta^e)^2 \leq \frac{1}{\omega^e} g(b_\gamma) \rightarrow 0.$$

Thus the residual means in all observed environments with positive weight converge to zero.

Now let $v \sim N(\delta, \Omega)$ with

$$N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}.$$

By definition of the mean-only Gaussian uncertainty class,

$$\delta \delta^\top \preceq \gamma M_{\text{obs}}.$$

Therefore

$$|w_\gamma^\top \delta|^2 = w_\gamma^\top \delta \delta^\top w_\gamma \leq \gamma w_\gamma^\top M_{\text{obs}} w_\gamma \rightarrow 0.$$

Since this bound holds for every admissible δ , we obtain the uniform convergence

$$\sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} |w_\gamma^\top \delta| \rightarrow 0.$$

It remains to translate these residual statements into statements about score distributions. Since $b_\gamma \rightarrow b_\infty^{\text{mean}}$, we also have $w_\gamma \rightarrow w_{b_\infty^{\text{mean}}}$. Hence the common residual variance in the mean-only setting satisfies

$$s_\gamma^2 := w_\gamma^\top \Omega w_\gamma \rightarrow s_\infty^2 := w_{b_\infty^{\text{mean}}}^\top \Omega w_{b_\infty^{\text{mean}}}.$$

Moreover, $s_\infty^2 > 0$, because $\Omega \succ 0$ and

$$w_{b_\infty^{\text{mean}}} = K^\top \begin{pmatrix} -b_\infty^{\text{mean}} \\ 1 \end{pmatrix} \neq 0.$$

Let $Q_\infty := \mathcal{L}(|N(0, s_\infty^2)|)$ denote the centered folded-normal distribution with variance s_∞^2 . For an observed environment e , the score distribution is

$$P_{e,\gamma}^T = \mathcal{L}(T_\gamma^e) = \mathcal{L}\left(|N(w_\gamma^\top \delta^e, s_\gamma^2)|\right).$$

Since $w_\gamma^\top \delta^e \rightarrow 0$ and $s_\gamma^2 \rightarrow s_\infty^2 > 0$, the Gaussian residual distributions converge in total variation to $N(0, s_\infty^2)$. Applying the map $u \mapsto |u|$ cannot increase total variation distance, so

$$P_{e,\gamma}^T \rightarrow Q_\infty$$

in total variation for every observed environment e with $\omega^e > 0$.

The calibration score distribution is the weighted mixture

$$P_{\text{cal},\gamma}^T = \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e P_{e,\gamma}^T.$$

Therefore,

$$\begin{aligned} 2 \|P_{\text{cal},\gamma}^T - Q_\infty\|_{\text{TV}} &= 2 \left\| \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (P_{e,\gamma}^T - Q_\infty) \right\|_{\text{TV}} \\ &\leq \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e 2 \|P_{e,\gamma}^T - Q_\infty\|_{\text{TV}} \rightarrow 0. \end{aligned}$$

Observed environments with zero weight do not contribute to the mixture. Thus

$$P_{\text{cal},\gamma}^T \rightarrow Q_\infty$$

in total variation.

For an admissible test distribution $N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}$, the test residual distribution is

$$w_\gamma^\top v \sim N(w_\gamma^\top \delta, s_\gamma^2).$$

The uniform bound above shows that

$$\sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} |w_\gamma^\top \delta| \rightarrow 0,$$

and we already know that $s_\gamma^2 \rightarrow s_\infty^2 > 0$. By continuity of Gaussian laws in total variation with respect to their mean and variance parameters, and by the uniform bound on the projected means, the corresponding test score distributions converge uniformly in total variation to Q_∞ :

$$\sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - Q_\infty \right\|_{\text{TV}} \rightarrow 0.$$

Finally, by the triangle inequality,

$$\begin{aligned} &\sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - P_{\text{cal},\gamma}^T \right\|_{\text{TV}} \\ &\leq \sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - Q_\infty \right\|_{\text{TV}} + 2 \left\| Q_\infty - P_{\text{cal},\gamma}^T \right\|_{\text{TV}}. \end{aligned}$$

Both terms on the right-hand side converge to zero. Therefore,

$$\sup_{N(\delta, \Omega) \in \mathcal{C}_\gamma^{\text{mean}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - P_{\text{cal},\gamma}^T \right\|_{\text{TV}} \rightarrow 0.$$

This completes the proof.

A.7 Proof of Proposition 3.10

In the covariance-only setting, the observed environments satisfy

$$\xi^e \sim N(\delta, \Omega_e), \quad \Omega_e = \Omega_0 + H_e,$$

where the mean δ is common to all environments, $H_0 = 0$, and $H_e \succeq 0$ for all $e \in \mathcal{E}_{\text{obs}}$. Recall that

$$\Delta_{\text{obs}} = \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e H_e.$$

The residual of a linear predictor b in environment e satisfies

$$U^e(b) = w_b^\top \xi^e \sim N(w_b^\top \delta, w_b^\top \Omega_e w_b).$$

In the baseline environment,

$$R_0(b) = \mathbb{E}[U^0(b)^2] = w_b^\top (\Omega_0 + \delta \delta^\top) w_b.$$

For an observed environment e ,

$$\begin{aligned} R_e(b) &= w_b^\top (\Omega_e + \delta \delta^\top) w_b \\ &= w_b^\top (\Omega_0 + \delta \delta^\top) w_b + w_b^\top H_e w_b. \end{aligned}$$

Thus the fixed-reference objective becomes

$$\begin{aligned} L_\gamma^{\text{ref}}(b) &= R_0(b) + \gamma \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (R_e(b) - R_0(b)) \\ &= w_b^\top (\Omega_0 + \delta \delta^\top) w_b + \gamma w_b^\top \Delta_{\text{obs}} w_b \\ &= a(b) + \gamma g(b), \end{aligned}$$

where

$$a(b) := w_b^\top (\Omega_0 + \delta \delta^\top) w_b, \quad g(b) := w_b^\top \Delta_{\text{obs}} w_b.$$

The function g is a nonnegative quadratic function, because $\Delta_{\text{obs}} \succeq 0$. Moreover, a is a strictly convex quadratic function of b . Indeed, $\Omega_0 + \delta \delta^\top \succ 0$, hence $K(\Omega_0 + \delta \delta^\top)K^\top \succ 0$, and the Hessian of a with respect to b is twice the upper-left $p \times p$ block of this positive definite matrix. In particular, $a(b) \rightarrow \infty$ whenever $\|b\| \rightarrow \infty$.

By the assumption of the proposition,

$$\mathcal{N}_{\text{cov}} = \{b \in \mathbb{R}^p : g(b) = 0\}$$

is nonempty, and

$$b_\infty^{\text{cov}} = \underset{b \in \mathcal{N}_{\text{cov}}}{\text{argmin}} a(b).$$

Thus Lemma 3.8 applies. Hence

$$b_\gamma \rightarrow b_\infty^{\text{cov}}$$

and

$$\gamma g(b_\gamma) = \gamma w_\gamma^\top \Delta_{\text{obs}} w_\gamma \rightarrow 0.$$

We now use this rate to control the observed projected variances and the admissible test projected variances. For any observed environment e with $\omega^e > 0$, the assumption $H_e \succeq 0$ gives

$$g(b_\gamma) = w_\gamma^\top \Delta_{\text{obs}} w_\gamma = \sum_{e' \in \mathcal{E}_{\text{obs}}} \omega^{e'} w_\gamma^\top H_{e'} w_\gamma \geq \omega^e w_\gamma^\top H_e w_\gamma.$$

Hence

$$0 \leq w_\gamma^\top H_e w_\gamma \leq \frac{1}{\omega^e} g(b_\gamma) \rightarrow 0.$$

Thus the additional projected variance caused by each observed covariance perturbation converges to zero.

Now let

$$v \sim N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}.$$

By definition of the covariance-only Gaussian uncertainty class,

$$0 \preceq H \preceq \gamma \Delta_{\text{obs}}.$$

Therefore

$$0 \leq w_\gamma^\top H w_\gamma \leq \gamma w_\gamma^\top \Delta_{\text{obs}} w_\gamma \rightarrow 0.$$

Since this bound holds for every admissible H , we obtain the uniform convergence

$$\sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} w_\gamma^\top H w_\gamma \rightarrow 0.$$

It remains to translate these residual statements into statements about score distributions. Since $b_\gamma \rightarrow b_\infty^{\text{cov}}$, we also have $w_\gamma \rightarrow w_{b_\infty^{\text{cov}}}$. Hence the baseline projected residual variance satisfies

$$s_{\gamma,0}^2 := w_\gamma^\top \Omega_0 w_\gamma \rightarrow s_\infty^2 := w_{b_\infty^{\text{cov}}}^\top \Omega_0 w_{b_\infty^{\text{cov}}}.$$

Moreover, $s_\infty^2 > 0$, because $\Omega_0 \succ 0$ and

$$w_{b_\infty^{\text{cov}}} = K^\top \begin{pmatrix} -b_\infty^{\text{cov}} \\ 1 \end{pmatrix} \neq 0.$$

The residual mean is common across all covariance-only shifts and equals

$$m_\gamma := w_\gamma^\top \delta.$$

Since $w_\gamma \rightarrow w_{b_\infty^{\text{cov}}}$, we have

$$m_\gamma \rightarrow m_\infty := w_{b_\infty^{\text{cov}}}^\top \delta.$$

Let

$$Q_\infty := \mathcal{L}(|N(m_\infty, s_\infty^2)|)$$

denote the limiting folded-normal distribution. For an observed environment e , the score distribution is

$$P_{e,\gamma}^T = \mathcal{L}(T_\gamma^e) = \mathcal{L}\left(|N(m_\gamma, s_{\gamma,0}^2 + w_\gamma^\top H_e w_\gamma)|\right).$$

Since

$$m_\gamma \rightarrow m_\infty, \quad s_{\gamma,0}^2 \rightarrow s_\infty^2, \quad w_\gamma^\top H_e w_\gamma \rightarrow 0,$$

the Gaussian residual distributions converge in total variation to $N(m_\infty, s_\infty^2)$. Applying the map $u \mapsto |u|$ cannot increase total variation distance, so

$$P_{e,\gamma}^T \rightarrow Q_\infty$$

in total variation for every observed environment e with $\omega^e > 0$.

The calibration score distribution is the weighted mixture

$$P_{\text{cal},\gamma}^T = \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e P_{e,\gamma}^T.$$

Therefore,

$$\begin{aligned} 2 \left\| P_{\text{cal},\gamma}^T - Q_\infty \right\|_{\text{TV}} &= 2 \left\| \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e (P_{e,\gamma}^T - Q_\infty) \right\|_{\text{TV}} \\ &\leq \sum_{e \in \mathcal{E}_{\text{obs}}} \omega^e 2 \left\| P_{e,\gamma}^T - Q_\infty \right\|_{\text{TV}} \rightarrow 0. \end{aligned}$$

Observed environments with zero weight do not contribute to the mixture. Thus

$$P_{\text{cal},\gamma}^T \rightarrow Q_\infty$$

in total variation.

For an admissible test distribution

$$N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G},$$

the test residual distribution is

$$w_\gamma^\top v \sim N(m_\gamma, s_{\gamma,0}^2 + w_\gamma^\top H w_\gamma).$$

The uniform bound above shows that

$$\sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} w_\gamma^\top H w_\gamma \rightarrow 0,$$

and we already know that

$$m_\gamma \rightarrow m_\infty, \quad s_{\gamma,0}^2 \rightarrow s_\infty^2 > 0.$$

By continuity of Gaussian laws in total variation with respect to their mean and variance parameters, and by the uniform bound on the projected variance perturbations, the corresponding test score distributions converge uniformly in total variation to Q_∞ :

$$\sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - Q_\infty \right\|_{\text{TV}} \rightarrow 0.$$

Finally, by the triangle inequality,

$$\begin{aligned} &\sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - P_{\text{cal},\gamma}^T \right\|_{\text{TV}} \\ &\leq \sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - Q_\infty \right\|_{\text{TV}} + 2 \left\| Q_\infty - P_{\text{cal},\gamma}^T \right\|_{\text{TV}}. \end{aligned}$$

Both terms on the right-hand side converge to zero. Therefore,

$$\sup_{N(\delta, \Omega_0 + H) \in \mathcal{C}_\gamma^{\text{cov}, G}} 2 \left\| \mathcal{L}(|w_\gamma^\top v|) - P_{\text{cal},\gamma}^T \right\|_{\text{TV}} \rightarrow 0.$$

This completes the proof.



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Place, date

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¹ For further information please consult the ETH Zurich websites, e.g. <https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html> and <https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html> (subject to change).